

Optimising covariate allocation at design stage using Fisher Information Matrix for Non-Linear Mixed Effects Models in pharmacometrics

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Abstract

This work focuses on designing experiments for pharmacometrics studies analysed by Non-Linear Mixed Effects Models (NLMEM) including covariates to describe inter-individual variability. Covariate effects may help identify patient subpopulations at risk of sub-therapeutic or toxic responses, for instance when hepatic or renal impairment reduces drug elimination, increasing safety risks. Before collecting and modelling new clinical trial data, choosing an appropriate design is crucial, particularly to ensure sufficient information to estimate covariate effects and their uncertainty. Assuming a known NLMEM with covariate effects and a joint distribution for covariates in the target population from previous clinical studies, we propose to optimise the allocation of covariates among the subjects to be included in a new trial. It aims achieving better overall parameter estimations and therefore increase power of statistical tests on covariate effects to detect the clinical relevance or non-relevance of relationships. We used the Fisher Information Matrix and developed a fast and deterministic computation method, leveraging Gaussian quadrature and copula modelling. We suggested dividing the domain of continuous covariates into clinically meaningful intervals and optimised their proportions, along with the proportion of each category for the discrete covariates. The optimisation problem was formulated as a convex problem subject to linear constraints, allowing resolution using Projected Gradient Descent algorithm. Different scenarios for a pharmacokinetics model including either biological measurements of renal or hepatic function as covariate were explored. We showed that covariate optimisation reduces the number of subjects needed to achieve desired power in covariate tests for relevance or non-relevance.

Keywords:

Pharmacometrics, Optimal design, Fisher Information Matrix, Non-Linear Mixed Effects Models, Covariates, Sample size

Highlights

- Fisher Information Matrix in nonlinear mixed effects models (NLMEM) with covariates
- New computation method of Information Matrix using copula and Gaussian quadrature
- Optimisation of covariate allocation using partitioning and projected gradient
- Reduced sample size needed to detect clinically relevant covariate in a PK example

1. Introduction

In pharmacometrics longitudinal data collected during clinical trials are analysed through Non-Linear Mixed Effect Models (NLMEM). Notably, NLMEM allow to quantify the relationships between covariates and parameters (Mould and Upton, 2012). Indeed, covariates are often included in pharmacometrics models to describe predictable between inter-individual variability (IIV), thereby enhancing model fit and/or model-based predictions (Sanghavi et al., 2024). Covariates are individual-specific characteristics known at baseline, which may be either discrete (e.g., sex, ethnicity) or continuous (e.g., body weight, creatinine clearance). In addition, covariate effects may help identify patient subpopulations at risk of sub-therapeutic or toxic responses, thus supporting the individualization of drug therapy (Joerger, 2012). Especially, in pharmacokinetics (PK) covariates can have effects on drug exposure requiring dose adjustment whose clinical relevance must be proven (FDA, 2022). This is particularly true for renal or hepatic impairment which can reduce the elimination of drugs and lead to increased safety risks (FDA, 2024, 2003; EMA, 2016, 2005). These dysfunctions can be quantified using one or more correlated biological covariates. When clinical studies enrolled a diverse patient population covering the full range of covariate values, and when sufficient PK data are available to evaluate the impact of organ impairment, population PK analyses from phase 2 and/or phase 3 trials may be sufficient to estimate the effect of renal function (RF) and/or hepatic function (HF) on drug exposure and its uncertainty within the studied population (FDA, 2024). Otherwise, dedicated additional studies should be conducted. Regulatory guidance recommends the use of confidence intervals to interpret change of exposure as ratios between groups (FDA, 2024, 2003; EMA, 2016, 2005), thereby supporting a quantitative assessment of potential PK alterations.

Our objective is therefore to support the design of future clinical trials in pharmacometrics by computing optimal covariate distributions that ensure sufficient statistical power to detect clinically relevant or non-relevant covariate relationships. In particular, we aim to determine the required number of subjects within subpopulations of interest, and thus avoid conducting specific trials to assess the effect of specific impairments on PK.

Before collecting and modelling the data, a key step is indeed to choose an appropriate design. In NLMEM, a population design refers to the number of subjects and their allocation to elementary designs, an elementary design being a combination of values for the design variables, such as number of samples, sampling times, dosage regimen, etc. The aim of the design phase is thus to improve the structure of a study to maximize the informativeness of the data that would be collected. This approach is particularly valuable when dealing with sparse sampling (e.g., paediatrics, rare diseases), ethical or logistical constraints or high IIV. Usually, covariates are not considered at this step, and although it is commonly recommended to include subjects spanning the full range of covariate values (FDA, 2022; Sanghavi et al., 2024), there is currently no method for quantifying the influence of their distribution within the range. In its Population PK Guidance for Industry 2022 (FDA, 2022), the FDA emphasises that "Simulations and optimal design methods can maximize the utility of population PK data collection and analyses" and stresses in particular that these methods can aim to estimate "major covariate effects of interest" with greater precision. Nevertheless, the method proposed for power assessment and sample size computation in this guidance is clinical trial simulations (CTS), although it is computationally expensive and less exhaustive than optimal design strategies.

The traditional theory of optimal experimental design, initially developed for non-linear models (Atkinson et al., 2007), have been imported to the field of NLMEM and pharmacometrics (Mentre et al., 1997; Mentre et al., 2013; Strömberg et al., 2016) and relies on Fisher Information Matrix

(FIM) computation. In the NLMEM framework, the FIM does not have a closed-form solution, requiring the computation of two intractable integrals: the integration of the derivatives of the log-likelihood over the random effects, and the evaluation of their expectation with respect to the observations. For continuous responses, a first-order linearisation of the model around 0, the mean of the random effects, results in a Gaussian approximation for which an analytical formula can be computed (Mentre et al., 1997; Bazzoli et al., 2009). Otherwise, the expectation over the observations can be approximated through a Monte Carlo procedure coupled with a linearisation of the model around the empirical Bayes estimate of the random effects, to bypass the integration over the random effects (Retout and Mentré, 2003). For design purpose with continuous responses, the linear approximation has been shown to be appropriate (Strömberg et al., 2016). If the response is discrete or if a linear approximation is not appropriate, other methods have been proposed, such as Markov Chain Monte Carlo (MCMC) to integrate over the random effects (Riviere et al., 2016), or Adaptive Gaussian Quadrature (for continuous (Nguyen and Mentré, 2014) or discrete (Ueckert and Mentré, 2017) outcomes), both of which combine Monte Carlo methods to evaluate the expectation over the observations. In those works, covariates were generally either absent or fixed, except in (Retout and Mentré, 2003), where the integral over the covariate distribution was computed using a Monte Carlo procedure. Indeed, when covariates are treated as random variables, the FIM becomes an expectation over both the observations and the covariates. As a result, an additional integral must be computed with respect to the distribution of the covariate vector. The linearisation approach has been implemented in various software tools (Nyberg et al., 2015). Thus, before the start of a new study, given a NLMEM and parameters value, one can compute the expected standard error (SE) of the model parameters and deduce whether enough information will be collected to meet the objectives of the study. Thereafter, design optimisation consists in finding the design of experiments that provides the most informative data for estimating model parameters while respecting some design constraints, such as number of subjects, number of sampling times or samplings times windows.

Nonetheless, accounting for covariates and their distribution during the design phase enables the quantification of how both the variations in covariates and the design influence the confidence intervals of covariate effects and the statistical power of the associated tests. It also enables assessment of the impact of covariate distributions on parameter identifiability and the determination of the minimum sample size required to achieve a desired statistical power. However, this aspect has not been adequately addressed in the literature, and existing software tools handle only known and fixed covariate values, rather than full covariate distributions. Previous works have looked at how to account for covariates when allocating patients to different arms, and how an unbalanced design could improve efficiency and ethics, including model-based approaches (see for instance (Rosenberger and Sverdlov, 2008) for a review). Nevertheless, these approaches do not use the longitudinal data framework nor NLMEM. In addition, they aim to optimise the design sequentially and seek, for each new patient entering the trial, to allocate them to the best possible arm in terms of efficacy and ethical considerations, while maintaining randomisation. Consequently, it does not correspond to our setting aiming to optimise the design prior to setting up the trial. In (Bogacka et al., 2017), a new method based of the FIM for optimising design using NLMEM where covariates are design variables is proposed. However, the proposed methods require the FIM to be evaluated for a known set of covariate values, without taking into account the distribution of these covariates, so the combinatorial approach can result in a very large number of elementary designs to be evaluated, and the optimisation result is discrete, making it difficult to manage continuous covariates.

In a previous work, we explored methods for handling covariates distribution in NLMEM with

FIM computed by linearisation (Fayette et al., 2025). The integral with respect to covariate distribution was approximated using Monte Carlo integration, and we proposed to leverage either an historical covariate dataset, a known covariate distribution or a copula. This approach enables simultaneous handling of both discrete and continuous covariates, while accounting for their correlations. The accuracy in predicting uncertainty and power of significance and relevance tests was assessed in a simulation study. This approach was also used to optimize the sampling scheme in an in silico study comparing imputation methods for missing data (Duflot et al., 2025).

In the present work, we developed a new method aimed at reducing the computational cost of the integration with respect to covariate distribution in the linearised FIM. Indeed, with optimisation purposes, the required number of FIM evaluation increases and run time becomes a challenge. Therefore, avoiding Monte Carlo simulations is a lever to reduce this cost. As a consequence, we introduce a Gaussian quadrature (Golub and Welsch, 1969) (GQ) based approach, thus faster and deterministic. We also developed a workflow to determine the number of quadrature nodes required to ensure accurate FIM computation. We have coupled this approach with the use of copula to describe the distribution of covariates, as they are very flexible and, if already available, can be used without individual data.

Regarding optimisation, a hurdle to be overcome is that optimising the distribution of continuous covariates is not trivial as it is unrealistic to recommend including a given number of patients matching an exact value of the covariate. Therefore, we propose to segment the continuous covariates into different meaningful intervals and to optimise the proportion of each of them among the trial population. The combination of possible continuous intervals and discrete categories defines a finite number of subgroups describing the entire population. This has the combined advantage to remain realistic in practice while defining discrete parameters to be optimised. Thus, the covariate distribution in the population is here defined as a mixture of subgroups distributions, and we propose a method to construct an enrolment strategy based on the weighting of these subgroups. The FIM is computed over a finite set combining elementary designs and covariate distributions and each combination is optimally weighted, given the constraints, using the Projected Gradient Descent (PGD) algorithm (Calamai and Moré, 1987). Once the covariate distribution has been optimised, the sample size can be estimated to achieve the desired level of precision, as well as the number of subjects needed (NSN) to attain the desired power for relevance tests on covariate effects.

Section 2 details the notations and methods developed in this work. In Section 3, we present an example based on a one-compartment PK model, exploring various scenarios to demonstrate the benefits of optimizing covariate distributions for detecting the relevance or non-relevance of covariate effects in the context of renal or hepatic impairment. Since prior data or copulas are not always available at the design stage, we used publicly available data from the National Health and Nutrition Examination Surveys (NHANES) database (Centers for Disease Control and Prevention (CDC).National Center for Health Statistics (NCHS), 2009-2020) to fit copula as prior covariate distributions. Otherwise, data available from previous studies may also be used. The results are presented in Section 4 and discussed in Section 5.

2. Methods

In this section we detail notations and methods on study design and NLMEM. We then describe the FIM integration using copula and Gauss-Legendre Quadrature (GLQ). Next, we present the optimisation of covariate distributions, with a particular focus on the segmentation of continuous covariates for optimisation purposes and how it is handled in the FIM computation. Finally, we in-

170 introduce the computation of ratios used to assess the clinical relevance or non-relevance of covariate
171 relationships.

172 2.1. Notations on study designs and NLMEM

173 In the following, a population design is denoted $\Xi = \{N, (\xi_1, \dots, \xi_N)\}$, where N is the number
174 of subjects and the ξ_i are their elementary designs.

175 The observation vector y_i for the i^{th} subject is modelled as given in equation (1) where f denotes
176 the structural non-linear model and $\theta_i = u(\mu, \beta, Z_i, \eta_i)$, the p -vector of individual parameters.
177 The latter are defined as a function u , depending on fixed effects, composed of the typical values
178 denoted μ and of the covariate effects β , and on individual random effects η_i , where $\eta_i \sim \mathcal{N}(0, \Omega)$.
179 Individual parameters also depend on the vector of possibly transformed individual covariates Z_i .

$$y_{ij} = f(\xi_i, \theta_i) + g(\xi_i, \theta_i, \sigma)\varepsilon_{ij} \quad (1)$$

180 The residual error is modelled by the function g , depending on ξ_i , θ_i and some parameters σ ,
181 and by the random variable ε_i , with $\varepsilon_i \sim \mathcal{N}(0, I_{n_i})$.

182 We denote $Z_i = (Z_{Dis,i}, Z_{Cont,i})$ a covariate vector when considered as a random vector, and
183 $z_i = (z_{Dis,i}, z_{Cont,i})$ as a realisation of this random vector. The lower $._{Dis}$ refers to the discrete sub-
184 vector of the covariate vector, while $._{Cont}$ refers to the continuous sub-vector. In pharmacometrics
185 modelling, parameters are generally log-normally distributed with additive covariate relationships
186 on the log scale, i.e. $\log \theta_i = \log \mu + \beta z_i + \eta_i$.

187 The P -vector of population parameters is denoted $\Psi = \{\mu, \beta, \lambda\}$, where λ contains the variance
188 parameters (elements of Ω and of the residual error model σ).

189 2.2. Elementary and population FIM

190 Given an elementary design ξ_i , a population parameters vector Ψ , and a covariate realisation
191 z_i the expected elementary FIM, $M_F(\xi_i, \Psi, z_i)$, is defined as the covariance matrix of the Fisher
192 score (equation (2)) where $l(y_i; \Psi, z_i, \xi_i)$ is the likelihood of the vector of observations y_i for the
193 population parameters Ψ , given the individual vector of covariates z_i and the elementary design ξ_i
194 (equation (3)).

$$M_F(\xi_i, \Psi, z_i) = \mathbb{E} \left(\frac{\partial \log l(y_i; \Psi, z_i, \xi_i)}{\partial \Psi} \frac{\partial \log l(y_i; \Psi, z_i, \xi_i)}{\partial \Psi^T} \right) \quad (2)$$

$$l(y_i; \Psi, z_i, \xi_i) = \int p(y_i | \eta_i; \Psi, z_i, \xi_i) p(\eta_i; \Psi, z_i) d\eta_i \quad (3)$$

196 Knowing the distribution p_Z for the covariate vector, the expected FIM can be expressed as an
197 expectation over the covariate vector as given in equation (4).

$$M_F(\xi_i, \Psi, p_Z) = \int M_F(\xi_i, \Psi, z) p_Z(z) dz \quad (4)$$

198 Assuming that the covariate distribution p_Z is the same for all the elementary design, the
199 expected population FIM is the sum of the elementary FIM (equation (5)).

$$M_F(\Xi, \Psi, p_Z) = \sum_{i=1}^N M_F(\xi_i, \Psi, p_Z) \quad (5)$$

200 In the following, for sake of simplicity we consider that the N subjects have the same elementary
201 design ξ , thus the expected population FIM is $M_F(\Xi, \Psi, p_Z) = N M_F(\xi, \Psi, p_Z)$.

In the case with only D discrete covariates, each with Q_δ , $\delta = 1, \dots, D$ categories. The covariate vector follows a discrete distribution with $n^{(QD)} = \prod_{\delta=1}^D Q_\delta$ possible values, denoted $z_{Dis,d}$, $d = 1, \dots, n^{(QD)}$, each having the probability $p_{Dis,d}$, $d = 1, \dots, n^{(QD)}$. Therefore, the elementary FIM for ξ writes as a weighted sum as given in equation (6).

$$M_F(\xi, \Psi, p_Z) = \sum_{d=1}^{n^{(QD)}} p_{Dis,d} M_F(\xi, \Psi, z_{Dis,d}) \quad (6)$$

In case there are both non-independent discrete and continuous covariates, for each of its possible value $z_{Dis,d}$, we denote $p_{Cont|z_{Dis,d}}$ the distribution of the continuous sub-vector conditionally to $z_{Dis,d}$. The elementary FIM for a given ξ is given in equation (7).

$$M_F(\xi, \Psi, p_Z) = \sum_{d=1}^{n^{(QD)}} p_{Dis,d} M_F(\xi, \Psi, z_{Dis,d}, p_{Cont|z_{Dis,d}}) \quad (7)$$

2.3. FIM integration using copula and Gauss-Legendre Quadrature

To compute the elementary FIM given in equation (7), the $n^{(QD)}$ matrices $M_F(\xi, \Psi, z_{Dis,d}, p_{Cont|z_{Dis,d}})$ must be evaluated. For the sake of simplicity, we omit the index d in the remainder of this paragraph. The first aspect of our method is to characterise the covariate distribution through copula modelling to account for correlations in FIM computation as already proposed (Fayette et al., 2025). Indeed, to perform GLQ integration, an explicit formula for the joint density of the covariates is required. We opted to use copulas rather than assuming a pre-specified multivariate distribution (e.g., multivariate Gaussian), due to their greater flexibility in capturing complex dependency structures.

It is well known that given a random vector $(Z_1, Z_2, \dots, Z_C)$ which follows the distribution $p_{Cont, |z_{Dis}}$ and has continuous marginals $F_1, \dots, F_C$, the random vector $(U_1, U_2, \dots, U_C) = (F_1(Z_1), F_2(Z_2), \dots, F_C(Z_C))$ has marginals uniformly distributed on $[0; 1]$. By definition, the copula of $(Z_1, Z_2, \dots, Z_C)$ is the cumulative distribution function (cdf) of $(U_1, U_2, \dots, U_C)$. The Rosenblatt transform (Rosenblatt, 1952) denoted $T_{Cont, |z_{Dis}}$ is defined in equation (8) and its inverse in equation (9).

$$T_{Cont, |z_{Dis}} : \begin{pmatrix} z_1 \\ \vdots \\ z_C \end{pmatrix} \mapsto \begin{pmatrix} \mathbb{P}(Z_1 \leq z_1) \\ \mathbb{P}(Z_2 \leq z_2 | Z_1 \leq z_1) \\ \vdots \\ \mathbb{P}(Z_C \leq z_C | Z_{C-1} \leq z_{C-1}, \dots, Z_1 \leq z_1) \end{pmatrix} = \begin{pmatrix} F_1(z_1) \\ F_2(z_2 | z_1) \\ \vdots \\ F_C(z_C | z_{C-1}, \dots, z_1) \end{pmatrix} \quad (8)$$

$$T_{Cont, |z_{Dis}}^{-1} : \begin{pmatrix} u_1 \\ \vdots \\ u_C \end{pmatrix} \mapsto \begin{pmatrix} F_1^{-1}(u_1) \\ F_2^{-1}(u_2 | u_1) \\ \vdots \\ F_C^{-1}(u_C | u_{C-1}, \dots, u_1) \end{pmatrix} \quad (9)$$

Therefore, applying the inverse Rosenblatt transform to a vector of random uniform variables return a vector that has the same distribution as $(Z_1, Z_2, \dots, Z_C)$. Consequently, $M_F(\xi, \Psi, z_{Dis}, p_{Cont, |z_{Dis}})$ can be expressed using $T_{Cont, |z_{Dis}}^{-1}$ as given in equation (10).

$$M_F(\xi, \Psi, z_{Dis}, p_{Cont, |z_{Dis}}) = \int_{[0;1]^C} M_F(\xi, \Psi, z_{Dis}, T_{Cont, |z_{Dis}}^{-1}(u_1, \dots, u_C)) du_1 \dots du_C \quad (10)$$

The second aspect is to approximate this integral through GQ (Golub and Welsch, 1969). The latter consists in approximating an integral by a weighted sum of the function to be integrated, evaluated at a number $n^{(Q)}$ of given points in the integration domain, called quadrature nodes. The advantage of GQ integration over the Monte Carlo is that, since the nodes are chosen to represent the distribution harmoniously, equivalent accuracy is achieved with fewer GQ nodes than with random draws for Monte Carlo (James, 1980).

In our case the integration domain for each of the C integrals is $[0; 1]$. Therefore, we apply C GLQ with a substitution, as GLQ allows to integrate over the interval $[-1, 1]$. Denoting v_q and w_q the q^{th} node and its weight in GLQ, the substitutions are: $v'_q = \frac{1}{2}(1 + v_q)$ and $w'_q = \frac{w_q}{2}$. Finally, $M_F(\xi, \Psi, z_{Dis}, p_{Cont}, |z_{Dis})$ can be expressed as given in equation (11).

$$M_F(\xi, \Psi, z_{Dis}, p_{Cont}, |z_{Dis}) \approx \sum_{q_1=1}^{n^{(Q)}} \cdots \sum_{q_C=1}^{n^{(Q)}} w'_{q_1} \dots w'_{q_C} M_F(\xi, \Psi, z_{Dis}, T^{-1}(v'_{q_1}, \dots, v'_{q_C})) \quad (11)$$

2.4. Optimisation of covariate distributions

We assume here that initial distributions are known for the covariates, and our aim is to optimally allocate subjects within the study accordingly. In this subsection, we describe first the segmentation of continuous covariates for optimisation, then the optimisation criteria, the proposed optimisation algorithm and the derivation of NSN .

2.4.1. Segmentation of continuous covariates

In order to optimise the allocation for continuous covariates, we suggest segmenting the continuous covariates into different intervals, chosen for their clinical meaning. We then distribute the allocation between the intervals of continuous covariates and between the categories of discrete covariates. The advantage is that it preserves the continuous nature of the covariates in the FIM calculus, while providing a discrete segmentation. Such segmentation is advantageous for reducing dimensionality during optimisation and for guiding subject inclusion based on the optimisation results.

We consider a general framework, with respectively D discrete and C continuous covariates. The domain of each continuous covariate, indexed by c , is segmented in $n_c^{(I)}$ intervals denoted $\mathcal{I}_j^{(c)}$, $j = 1, \dots, n_c^{(I)}$. Combining those continuous intervals leads to $n^{(I)} = \prod_c n_c^{(I)}$ possible intervals for Z_{Cont} . Finally, there are $L = n^{(QD)} \times n^{(I)}$ possible distributions for the covariate vector Z . Conditionally to $Z_{Dis} = z_{Dis,d}$, the distribution of $Z_{Cont} = (Z_1 \in \mathcal{I}_{j_{c_1}}^{(c_1)}, \dots, Z_C \in \mathcal{I}_{j_{c_C}}^{(c_C)})$ is denoted $p_{Cont, |z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})}$. The proportion of subjects within the trial population and for which $Z_{Dis,i} = z_{Dis,d}$ and $Z_{Cont,i} \sim p_{Cont, |z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})}$ is denoted: $x_{|z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})}$. This segmentation leads to the elementary FIM formula given in equation (12), which now depends on $\mathbf{x}$.

$$M_F(\xi, \Psi, p_Z(\mathbf{x})) = \sum_{d=1}^{n^{(QD)}} \sum_{i_1=1}^{n_{c_1}^{(I)}} \cdots \sum_{i_{nC}=1}^{n_{c_C}^{(I)}} x_{|z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})} M_F(\xi, \Psi, z_{Dis,d}, p_{Cont, |z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})}) \quad (12)$$

We denote $\mathbf{x}$ the vector of size L containing the $x_{|z_{Dis,d}, \mathcal{I}(j_{c_1}, \dots, j_{c_C})}$ and for a better clarity in the notations, we denote x_l the l^{th} element of $\mathbf{x}$ and $p_{Z,l}$ the associated distribution for Z . The overall

261 mixture distribution for Z is denoted $p_Z(\mathbf{x})$. These notations result in the FIM formula given in
 262 equation (13). Finally, the multivariate distribution of covariates within the trial is described as
 263 the mixture of L multivariate distributions, and it is these mixture weights that we are seeking to
 264 optimise.

$$M_F(\xi, \Psi, p_Z(\mathbf{x})) = \sum_{l=1}^L x_l M_F(\xi, \Psi, p_{Z,l}) \quad (13)$$

265 Each of the L matrices that appears in equation (13) can be computed using the GLQ approach
 266 presented above.

267 2.4.2. Optimisation criterion

268 In order to compare the FIM between different designs, a criteria mapping the matrix to a scalar
 269 needs to be defined. Among various optimality criterion used in theory of optimal design (see for
 270 instance (Atkinson et al., 2007)), the D-criterion is a popular one and is widely used in the field of
 271 pharmacometrics. It uses the determinant of the FIM, as given in equation (14) and maximizing
 272 it ensures that the parameter estimates are as precise as possible. Indeed, a larger determinant of
 273 the FIM corresponds to a smaller volume of the confidence ellipsoid for the parameter estimates,
 274 indicating higher precision.

$$\phi_D(\Xi, \Psi, p_Z(\mathbf{x})) = \left(\det(M_F(\Xi, \Psi, p_Z(\mathbf{x}))) \right)^{\frac{1}{P}} \quad (14)$$

275 The comparison of covariate repartition $\mathbf{x}$ against a reference $\mathbf{x}_{ref}$ for the D-criterion is done
 276 by calculating the D-efficiency E_D (equation (15)).

$$E_D(\Xi, \Psi, p_Z(\mathbf{x}_{ref}), \mathbf{x}) = \frac{\phi_D(\Xi, \Psi, p_Z(\mathbf{x}))}{\phi_D(\Xi, \Psi, p_Z(\mathbf{x}_{ref}))} \quad (15)$$

277 2.4.3. Optimising covariate proportions using Projected Gradient Descent

278 Our aim is to maximize the D-criterion, for a fixed design Ξ by distributing the various covariate
 279 distributions weighted by the x_l , and given some constraints on these proportions.

280 *Linear constraints.* First, as $\mathbf{x}$ represents a vector of proportions, it should sum to 1 and each x_l
 281 has to be in $[0; 1]$. Additional constraints can be set on the x_l , for instance to make sure that a
 282 given interval for a continuous covariate does not represent more than a desired proportion and
 283 not less than another proportion. Therefore there are both equality and inequality constraints,
 284 and the optimisation problem is given in equation (16). We denote $\mathcal{Q}$ the constraint space.

$$\begin{aligned} \min \quad & -\phi_D(\Xi, \Psi, p_Z(\mathbf{x})) \\ \text{s.t.} \quad & E\mathbf{x} = F \\ & G\mathbf{x} \geq H \end{aligned} \quad (16)$$

285 *Projected Gradient Descent.* As the D-criterion is a convex function, the PGD (Calamai and Moré,
 286 1987) can be used to solve the problem given in equation (16). Starting from an initial point $x_0 \in \mathcal{Q}$,
 287 PGD iterates the equation given in (17), where $\alpha_k \in [0; \infty]$ is the gradient step-size at the k^{th}

iteration, $\nabla\phi_D$ the gradient of ϕ_D with respect to $\mathbf{x}$ and $\mathcal{P}_Q$ is the projection on the constraint space. This projection is also the solution of a minimization problem, given in equation (18).

$$\mathbf{x}_{k+1} = \mathcal{P}_Q\left(\mathbf{x}_k + \alpha_k \nabla\phi_D(\Xi, \Psi, p_Z(\mathbf{x}_k))\right) \quad (17)$$

$$\mathcal{P}_Q(\mathbf{x}) = \min \|\mathbf{x} - \mathbf{y}\| \text{ s.t. } E\mathbf{y} = F, G\mathbf{y} \geq H \iff \mathcal{P}_Q(\mathbf{x}) = \min \frac{1}{2} \mathbf{y}^\top \mathbf{y} - \mathbf{x}^\top \mathbf{y} \text{ s.t. } E\mathbf{y} = F, G\mathbf{y} \geq H \quad (18)$$

PGD iterates until a stopping condition is met. The detailed computation for $\nabla\phi_D(\Xi, \Psi, p_Z, \mathbf{x}_k)$ when all the subjects have the same elementary design ξ is given in (Fayette et al., 2023) and the result is recall in equation (19), where tr denotes the trace operator.

$$\frac{\partial\phi_D(\Xi, \Psi, p_Z(\mathbf{x}))}{\partial x_l} = \frac{N}{P} \text{tr} \left(M_F(\Xi, \Psi, p_Z(\mathbf{x}))^{-1} M_F(\xi, \Psi, p_{Z,l}) \right) \det(M_F(\Xi, \Psi, p_Z(\mathbf{x})))^{1/P} \quad (19)$$

2.4.4. Sample size computation

After having optimised the proportion of each covariate distribution, we can compute the expected standard errors on the parameters as the square roots of the diagonal elements of the inverse of the FIM. From that, we can derive the sample size needed, NSN , all other things being equal, to reach desired confidence level on parameter estimates or on power of statistical tests on covariate effect. Denoting $SE(\Xi, \Psi, p_Z(\mathbf{x}))_\beta$ the standard error on a parameter $\beta \in \Psi$ for a design Ξ , population vector Ψ and a covariate distribution defined by p_Z and $\mathbf{x}$, the number of subjects needed NSN to reach a desired level SE^* on this parameter is given in equation (20).

$$NSN(\Xi, \Psi, p_Z(\mathbf{x}))_\beta = N \left(\frac{SE(\Xi, \Psi, p_Z(\mathbf{x}))_\beta}{SE^*} \right)^2 \quad (20)$$

From this, we can calculate the NSN required to achieve the desired power in covariate testing.

2.5. Evaluation of covariate relationships

Usually, to assess covariate relationships, significance test are performed to determine whether covariate effects are different from 0. The expected power for this test can be computed from the FIM (Retout et al., 2007; Fayette et al., 2025). But to assess the magnitude of a covariate effect, one can compute the ratio of change in primary parameters when covariate values change relative to a reference value z_{ref} . The uncertainty on this ratio can be derived from the SE on the covariate effect parameter. Then, the effect of the covariate value on the considered parameter will be said clinically relevant if the 90% confidence interval on the ratio is completely outside an equivalence interval $[R_{inf} ; R_{sup}]$ and clinically non-relevant if entirely within. If the confidence interval crosses the equivalence area, there is not enough power to conclude. Detailed formulas for ratios and powers are given in (Fayette et al., 2025). For a log-normally distributed parameter with an additive covariate relationship on the log scale, if the covariate is binary, the ratio writes $r_{l,c} = e^{\beta_{l,c}}$. If continuous, the ratio associated to a given percentile PX of the covariate distribution writes $r_{l,c(PX)} = e^{\beta_{l,c}(PX - P50)}$, where $P50$ denotes the median of the covariate distribution.

To assess the influence of belonging to an interval subset involving several covariates, we define the effect ratio as the expected value of the ratio function, conditional on the associated covariate sub-vector following the mixture distribution corresponding to the interval subset of interest.

Denoting by $\mathcal{L}^R$ the subset of covariate distributions defined over the interval subset, the ratio is given in Equation (21).

$$r_{\theta, \mathcal{L}^R} = \frac{1}{\sum_{l' \in \mathcal{L}^R} x_{l'}} \sum_{l \in \mathcal{L}^R} x_l \int r(\beta, z) p_{Z,l}(z) dz \quad (21)$$

The uncertainty associated with this composite ratio can be estimated using the Delta Method, and both relevance and non-relevance tests can be conducted based on the resulting Gaussian approximation.

3. Illustration of benefit of covariate distribution optimisation on a one-compartment PK model

In this section we explored a PK example to demonstrate the benefit of covariate distribution optimisation in reducing the sample size needed to draw conclusions through case studies, the first one in the context of renal impairment and the second of hepatic impairment.

3.1. PK example

3.1.1. Model

The PK model was a one compartment model with intravenous (IV) bolus and linear elimination, thus the drug concentration wrote $f(\xi_i = (t_i, Dose_i), \theta_i) = \frac{Dose_i}{V_i} e^{-\frac{Cl_i}{V_i} t_i}$, with two parameters: the clearance (Cl) and the volume of distribution (V), thus $\theta_i = (Cl_i, V_i)^\top$. The residual error was modelled as a combined error model: $g(\xi_i, \theta_i, \sigma = (a, b)^\top) = a + b f(\xi_i, \theta_i)$ and exponential random effects without correlation were assumed for V and Cl , $\eta_{V,i} \sim \mathcal{N}(0, \omega_V^2)$ and $\eta_{Cl,i} \sim \mathcal{N}(0, \omega_{Cl}^2)$. All covariate effects were modelled as additive on the log-scale. Both case studies include an effect of the Sex (SEX) and of the Body Mass Index (BMI) on V . BMI was first normalised and log transformed: $\log\left(\frac{BMI_i}{BMI_{pop}}\right)$, where BMI_{pop} denoted the reference BMI among reference individuals.

Renal case study. In the first case study an effect of RF on Cl was added through the creatinine clearance ($CLCR$), which was also normalised and log transformed: $\log\left(\frac{CLCR_i}{CLCR_{pop}}\right)$, where $CLCR_{pop}$ denoted the reference $CLCR$ among reference individuals.

Hepatic case study. In the second case study effect of HF was added on Cl , based on serum total Bilirubin ($BILI$) and serum Aspartate transaminase (AST) concentrations (Ramalingam et al., 2010), which were also normalised and log transformed: $\log\left(\frac{(BILI/ULN)_i}{(BILI/ULN)_{pop}}\right)$ and $\log\left(\frac{(AST/ULN)_i}{(AST/ULN)_{pop}}\right)$, where $(BILI/ULN)_{pop}$ and $(AST/ULN)_{pop}$ denote the references among reference individuals and ULN stands for Upper Limit Normal (ULN).

3.1.2. Covariate distribution specification

To reflect the case where covariates are not available at the design stage, we used the public database from the NHANES database (Centers for Disease Control and Prevention (CDC). National Center for Health Statistics (NCHS), 2009-2020) to get prior covariate distributions. Covariate data from 2009 to 2020 were extracted, keeping only subjects between 18 and 80 years old, with BMI higher or equal to 18.5 kg/m^2 and for whom there were no missing data for Age, SEX , Body Weight (BW), BMI , Height, Creatinine (CR_{serum}), $BILI$ and AST . 29 409 covariate vectors were selected in the NHANES database and Table 1 provides the summary statistics of the covariates.

The body weight was adjusted for ideal body weight computed with Robinson’s formula (Robinson et al., 1983) and the Cockcroft and Gault equation (equation (22)) with BW_{Adj} the adjusted body weight (Chin et al., 2013) was then used to estimate the creatinine clearance, $CLCR$.

$$CLCR = \frac{(140 - Age) \times BW_{Adj}}{72 \times CR_{serum}} \times 0.85^{SEX} \quad (22)$$

Both $BILI$ and AST were normalised by their respective ULN, which were 1.0 mg/dL for $BILI$ and 33 U/L for AST before 2017, and then 37 U/L for Male and 31 U/L for Female.

The BMI was segmented into three usual intervals, namely Obesity if $BMI \geq 30 \text{ kg/m}^2$, Overweight if $25 \leq BMI < 30 \text{ kg/m}^2$ and Healthy Weight if $18.5 \leq BMI < 25 \text{ kg/m}^2$. Of note, $BMI_{pop} = 22.8 \text{ kg/m}^2$, $CLCR_{pop} = 112.3 \text{ mL/min}$, $(BILI/ULN)_{pop} = 0.6$ and $(AST/ULN)_{pop} = 0.64$.

Renal case study. $CLCR$ was segmented into the four usual intervals to define RF, namely Normal RF if $CLCR \geq 90 \text{ mL/min}$, Mild RF if $60 \leq CLCR < 90 \text{ mL/min}$, Moderate RF if $30 \leq CLCR < 60 \text{ mL/min}$ and Severe RF if $CLCR < 30 \text{ mL/min}$. While there was a good balance between Male and Female and between the different BMI intervals, Severe RF was poorly represented with only 1.2% of the data. Combining SEX , BMI intervals and $CLCR$ intervals, there were $2 \times 4 \times 3 = 24$ possible covariate combinations. Nevertheless, those including Severe RF were poorly represented in the dataset (e.g. only 40 subjects are Male, with Severe RF and Obesity). Therefore, Severe RF individuals were pooled into two classes depending on their SEX and without taking into account their BMI status. This pooling was carried out to ensure the adequacy of copula fits. The detailed repartition between the resulting 20 covariate distributions is given in the first panel of Figure 8.

Hepatic case study. HF was characterized using the National Cancer Institute (NCI) Organ Dysfunction Working Group (ODWG) criteria (Ramalingam et al., 2010). According to these criteria, Normal HF is defined as $BILI/ULN \leq 1$ and $AST/ULN \leq 1$. Mild HF includes two subgroups: defined as $BILI/ULN \leq 1$ and $AST/ULN > 1$ or $1 < BILI/ULN \leq 1.5$ with any AST/ULN . For simplicity, we refer to these subpopulations as Mild1 and Mild2. Moderate and Severe HF are defined by $1.5 < BILI/ULN \leq 3$ and $BILI/ULN > 3$, respectively. Due to the very limited representation of Severe HF (only 10 subjects), Moderate and Severe HF were pooled, together accounting for only 1.2% of the covariate database. Consequently, as with RF, the ModerateSevere HF category was only combined with SEX . We chose to split the Mild HF category and model its distribution as a mixture of two copulas, each fitted independently to the Mild1 and Mild2 subgroups. As a result, a total of 20 copulas were fitted, while only 14 proportions required optimisation. The detailed distribution across the resulting 14 covariate distribution is shown in the first panel of Figure 11.

Reference individuals were defined as Male with Healthy Weight and both Normal RF and HF.

3.1.3. Parameter values

The PK parameter values and ratios of covariate effects are given in Table 2. With a ratio of 0.70, i.e. outside of $[0.80; 1.25]$, the effect of being a Female on V was tested for relevance; while $r_{V, \text{Obesity}}$ was within the equivalence interval, thus the effect of Obesity on V was tested for non-relevance. It should be noted that $r_{V, \text{Obesity}}$ was computed for the threshold value for belonging to this interval, namely for 30 kg/m^2 , in relation to the reference value, BMI_{pop} .

Table 1: Covariate data summary

Continuous covariates		
Covariate	mean (sd)	median [min; max]
<i>BMI</i> (kg/m ²)	29.5 (7)	28.2 [18.5; 92.3]
<i>CLCR</i> (ml/min)	99 (35.6)	97.3 [4.1; 342.9]
<i>BILI/ULN</i>	0.6 (0.3)	0.6 [0; 7.3]
<i>AST/ULN</i>	0.7 (0.5)	0.7 [0.2; 26.7]
Discrete covariates		
Covariate	Category Name	N (%)
<i>SEX</i>	Male	14 340 (48.8%)
	Female	15 069 (51.2%)
Continuous segmentation		
Condition on covariate	Interval Name	N (%)
$18.5 \leq BMI < 25 \text{ kg/m}^2$	Healthy Weight	8 217 (27.9%)
$25 \leq BMI < 30 \text{ kg/m}^2$	Overweight	9 620 (32.7%)
$BMI \geq 30 \text{ kg/m}^2$	Obesity	11 572 (39.3%)
$CLCR \geq 90 \text{ mL/min}$	Normal RF	17 187 (58.4%)
$60 \leq CLCR < 90 \text{ mL/min}$	Mild RF	8 327 (28.3%)
$30 \leq CLCR < 60 \text{ mL/min}$	Moderate RF	3 536 (12%)
$CLCR < 30 \text{ mL/min}$	Severe RF	359 (1.2%)
$BILI/ULN \leq 1$ and $AST/ULN \leq 1$	Normal HF	24 714 (84%)
$BILI/ULN \leq 1$ and $AST/ULN > 1$	Mild1 HF	2 597 (8.8%)
$1 < BILI/ULN \leq 1.5$	Mild2 HF	1 739 (5.9%)
$1.5 < BILI/ULN \leq 3$	Moderate HF	349 (1.2%)
$3 < BILI/ULN$	Severe HF	10 (0%)

BMI: refers to Body Mass Index, *CLCR*: Creatinine Clearance, *RF*: Renal function, *BILI*: Bilirubin, *AST*: Aspartate transaminase, *ULN*: Upper Limit Normal, *HF*: Hepatic function

Renal case study. The ratio assessing the impact of Severe RF on *Cl* was computed at the threshold value of 30 ml/min, relative to the reference $CLCR_{pop}$. The resulting ratio was 0.70, which lies outside the equivalence interval [0.80; 1.25], and was therefore tested for relevance.

A sensitivity analysis was conducted on this example to evaluate two aspects. First, the influence of IIV was assessed using two scenarios: a 'Normal IIV' scenario, with standard deviations as specified in Table 2, and a 'High IIV' scenario, in which these standard deviations were doubled. Second, the impact of covariate values was explored by varying $\beta_{Cl,CLCR}$, resulting in the ratio $r_{Cl, \text{Severe RF}}$ ranging from 0.40 to 0.75.

Hepatic case study. Through this case study we wanted to explore a non-relevant relationship, therefore small effect sizes were chosen with $\beta_{Cl,BILI/ULN} = 0.05$ and $\beta_{Cl,AST/ULN} = -0.38$. The ratio assessing the impact of ModerateSevere HF depends on both *BILI/ULN* and *AST/ULN*, and was computed as the expected value over covariate distributions involving ModerateSevere HF, as described in Section 2.5. The resulting ratio was 0.97, which falls within the equivalence range.

Table 2: Population PK parameter values used in the illustration example

Parameter	μ_{Cl} (L/h)	μ_V (L)	ω_{Cl}	ω_V	a (mg/L)	b
Value	0.43	7.13	0.24	0.30	2.39	0.08

Covariate effects

Parameter	$\beta_{V,Sex}$	$\beta_{V,BMI}$	$\beta_{Cl,CLCR}$	$\beta_{Cl,BILI/ULN}$	$\beta_{Cl,AST/ULN}$
Value	-0.35	0.50	0.27	0.05	-0.38

Ratios

Ratio	$r_{V, Female}$	$r_{V, Obesity}$	$r_{Cl, Severe RF}$	$r_{Cl, ModerateSevere HF}$
Value	0.70	1.14	0.70	0.97

μ .: refers to the typical value, β .: covariate parameter, ω : standard deviation of the random effect, a and b , parameters of the residual error model

$r_{l,c}$: ratio on the parameter l for the covariate c

3.1.4. Design

The study design includes 100 subjects, each receiving a 250 mg dose of the drug at time 0. For each subject, three blood samples are collected at 1, 4, and 12 hours post-dose. The concentration-time profiles, $f(t)$, accounting for univariate covariate effects and parameter values provided in Table 2, are shown in Figure 1. Because being Female reduces the volume of distribution, the typical concentration curve for Females lies above that for Males. Similarly, since a higher BMI increases the volume, the typical concentration curve for individuals with a Healthy Weight is higher than that for Overweight individuals, which in turn is higher than that for those with Obesity. Regarding RF, lower $CLCR$ values correspond to lower concentration curves, reflecting reduced clearance due to renal impairment. HF is more complex, as it depends on two covariates. When considered independently, higher $BILI/ULN$ values are associated with lower concentration curves, while higher AST/ULN values correspond to higher concentrations.

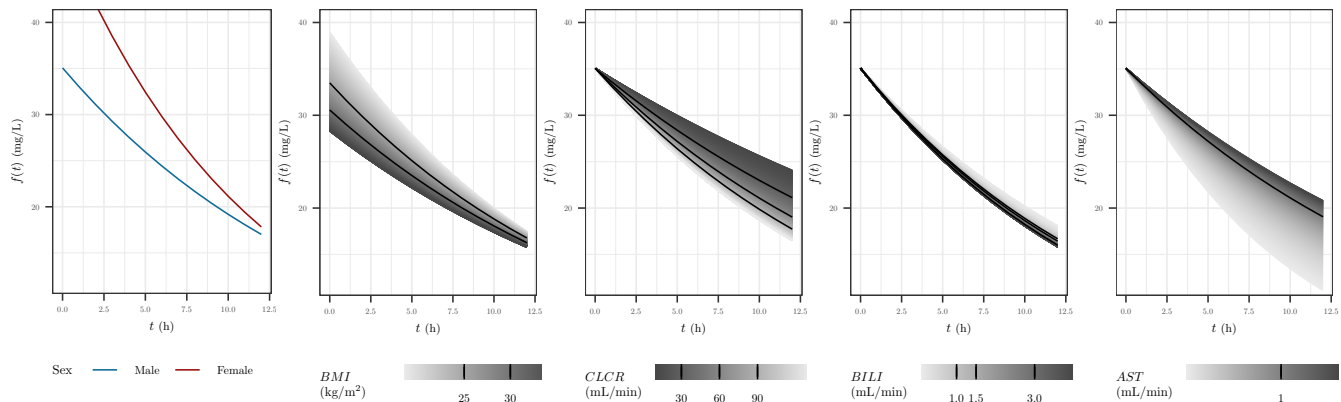


Figure 1: Evolution of the concentration $f(t)$ according to the fixed effects and univariate covariate effects: the first panel corresponds to the effect of SEX on V , the second panel BMI on V , the third panel $CLCR$ on Cl , the fourth panel $BILI/ULN$ on Cl and the fifth panel AST/ULN on Cl . For the last four panels, the black lines correspond to the limits of the intervals of the continuous covariates.

422 3.2. FIM evaluation using copula and quadrature

423 3.2.1. Copula fitting

424 In both case studies, all copula were fitted as vine copula, on the associated subset of data, using
 425 the R package *rvinecopulib* 0.6.3.1.1 (Nagler and Vatter, 2021). Copula fit evaluation was conducted
 426 through a simulation-based strategy as described in (Guo et al., 2024b) using 100 replicates. The
 427 relative error (RE) between summary metrics M_{sim} in copula-simulated populations and M_{obs}
 428 in the NHANES dataset was computed as $\frac{M_{sim} - M_{obs}}{M_{obs}}$. Summary metrics were the mean, the
 429 standard deviation, the median and the 5th and 95th percentiles of the marginal distribution of
 430 each continuous covariate. Overlaps between simulated and true distributions were also computed
 431 such as the simulated and observed correlations between continuous covariates. Diagnostics plots
 432 were explored using the R package *pmxcopula* (Guo et al., 2024a) with minor edits.

433 3.2.2. Quadrature settings

434 According to the GLQ rule, the approximation using $n^{(Q)}$ nodes is exact for integrands that are
 435 polynomials of degree $2n^{(Q)} - 1$ or less. Nevertheless, in our case, the integrand is the composition
 436 of the FIM and the inverse Rosenblatt transform of the copula describing the continuous covariate
 437 distribution, therefore, in a general setting, it is not a polynomial function. Consequently, the
 438 quadrature remains an approximation and increasing $n^{(Q)}$ increases the accuracy. To determine
 439 the number of nodes required in the quadrature for an acceptable accuracy in FIM evaluation, we
 440 computed the FIM starting with $n^{(Q)} = 1$ and going up to $n^{(Q)} = 15$, and looked for stabilization in
 441 the D-criterion by computing the relative difference in ϕ_D : $\Delta_D(n^{(Q)}) = 100 \frac{\phi_D^{(n^{(Q)})} - \phi_D^{(n^{(Q)}-1)}}{\phi_D^{(n^{(Q)}-1)}}$ between
 442 two number of nodes. We stopped when the latter was below 0.5%. The same number of nodes
 443 was used for all the continuous covariates and for each copula. As a target, we also computed
 444 the FIM using Monte Carlo simulations with $n^{(MC)} = 500$ Monte Carlo samples drawn from each
 445 copula and because of the stochasticity of Monte Carlo, we repeated this process 10 times and
 446 defined the target D-criterion as the average across the 10 repetitions.

447 3.3. Covariate distribution optimisation

448 In all scenarios, covariate distribution optimisation was carried using PGD. The step-size α was
 449 was fixed at 0.001. Optimisation was terminated when the change in the objective function between
 450 two successive iterations was less than $\epsilon = 10^{-5}$. Two constraint settings were considered. The
 451 first, referred to as 'Without constraints', imposed only the basic requirements that the proportions
 452 lie within the interval $[0, 1]$ and sum to 1. The second setting, referred to as 'With constraints',
 453 introduced additional constraints: each continuous interval was required to represent at least 5%
 454 of the population. Furthermore, the Severe RF and ModerateSevere HF were capped to 10% in
 455 each case study respectively. Indeed, such subjects are difficult to recruit in real-world practice.

456 3.4. Optimised distributions evaluation

457 The initial distribution was the observed proportions in the selected subset of the NHANES
 458 database. For each optimised distribution, the D-efficiency was calculated relative to the design
 459 based on the initial distribution. Additionally, the NSN to achieve 80% power for detecting the
 460 relevance or non-relevance of the relationships was computed.

461 3.5. Implementation

462 All the methods have been implementing using the R language (R Core Team, 2021). For
 463 FIM evaluation an extension of the package *PFIM*6.1 (Mentré et al., 2024), based on previous
 464 work (Fayette et al., 2025) have been done. All scripts are available online at [https://doi.org/](https://doi.org/10.5281/zenodo.14778033)
 465 [10.5281/zenodo.14778033](https://doi.org/10.5281/zenodo.14778033). The quadrature nodes were computed using the function *gauss.quad()*
 466 from the package *statmod* (Giner and Smyth, 2016). For PGD, the projection on the constraint
 467 space was made using the function *lsei()* from the package *limSolve* (Karline Soetaert et al., 2009).
 468 Roots for *NSN* were found using the R function *uniroot()* from the package *stats* (R Core Team,
 469 2013) which implements algorithms from (Brent, 2013).

470 4. Results

471 4.1. Copula fitting

472 The performance of copula model was evaluated for each covariate combination.

473 4.1.1. Renal case study

474 As shown in Appendix A, Figure A.13, RE was within $\pm 10\%$ for all the summary statistics for
 475 almost all the intervals of *BMI* and *CLCR*. The simulated correlations from the copula models
 476 were very similar to observed correlations between *BMI* and *CLCR* (Appendix A, Figure A.14)
 477 and overlap between simulated and observed join distribution for *BMI* and *CLCR* was always
 478 above 80%, except for Female with Severe RF, for which the overlap was above 80% in only in three
 479 quarters of the simulations (Appendix A, Figure A.15). Moreover, the VPC given in Figure 2
 480 showed a good match between simulated and observed level lines, for all combinations. Overall,
 481 copula fits were judged to be very satisfactory.

482 4.1.2. Hepatic case study

483 Regarding the hepatic case study, the overall fit quality appeared to be somewhat lower, partic-
 484 ularly in terms of marginal estimations for *AST/ULN*. Specifically, the relative error (Appendix
 485 A, Figure A.16) seemed to be less well controlled. Nevertheless, the correlation structure between
 486 variables was adequately captured (Appendix A, Figure A.14). The overlap (Appendix A, Fig-
 487 ure A.15) was not very satisfactory for the subgroup of females with ModerateSevere HF, which
 488 can be attributed to the limited number of observed subjects in the dataset (only 77). Importantly,
 489 however, the VPC (Figure 3) did not reveal any anomalies.

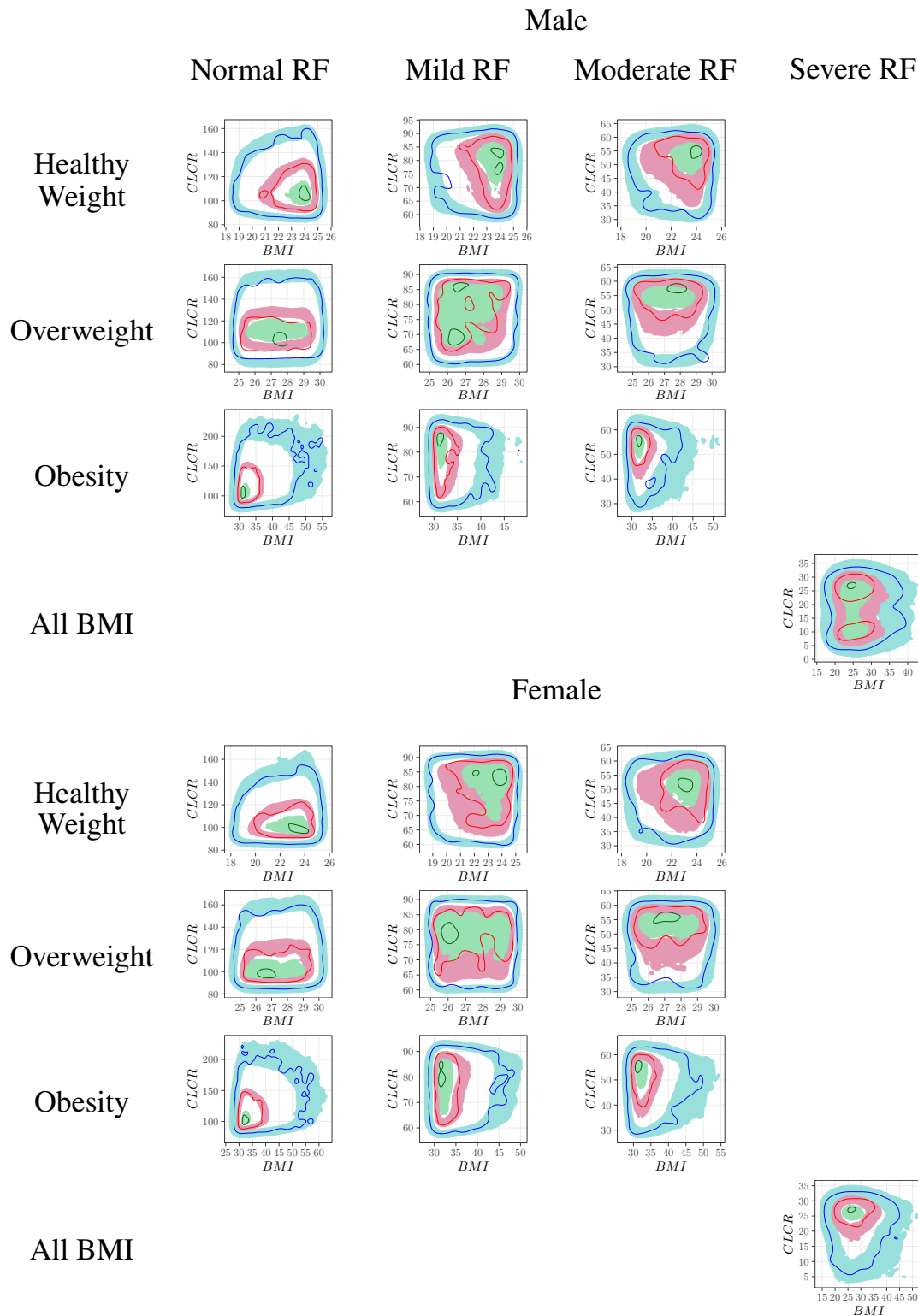


Figure 2: Renal case study - Visual predictive checks based on the contours of the bivariate density between BMI and $CLCR$. For each of the 20 covariate combination, the virtual population was simulated 100 times from the copula and the 99% prediction intervals of percentile contours of the joint distribution was derived and compared to the contours observed in the NHANES database.

The ribbon areas correspond to the 99% prediction interval of 5th (●), 50th (●) and 95th (●) percentile contours of the joint distribution of BMI and $CLCR$. The lines corresponds to the observed 5th (—), 50th (—) and 95th (—) percentile contours.

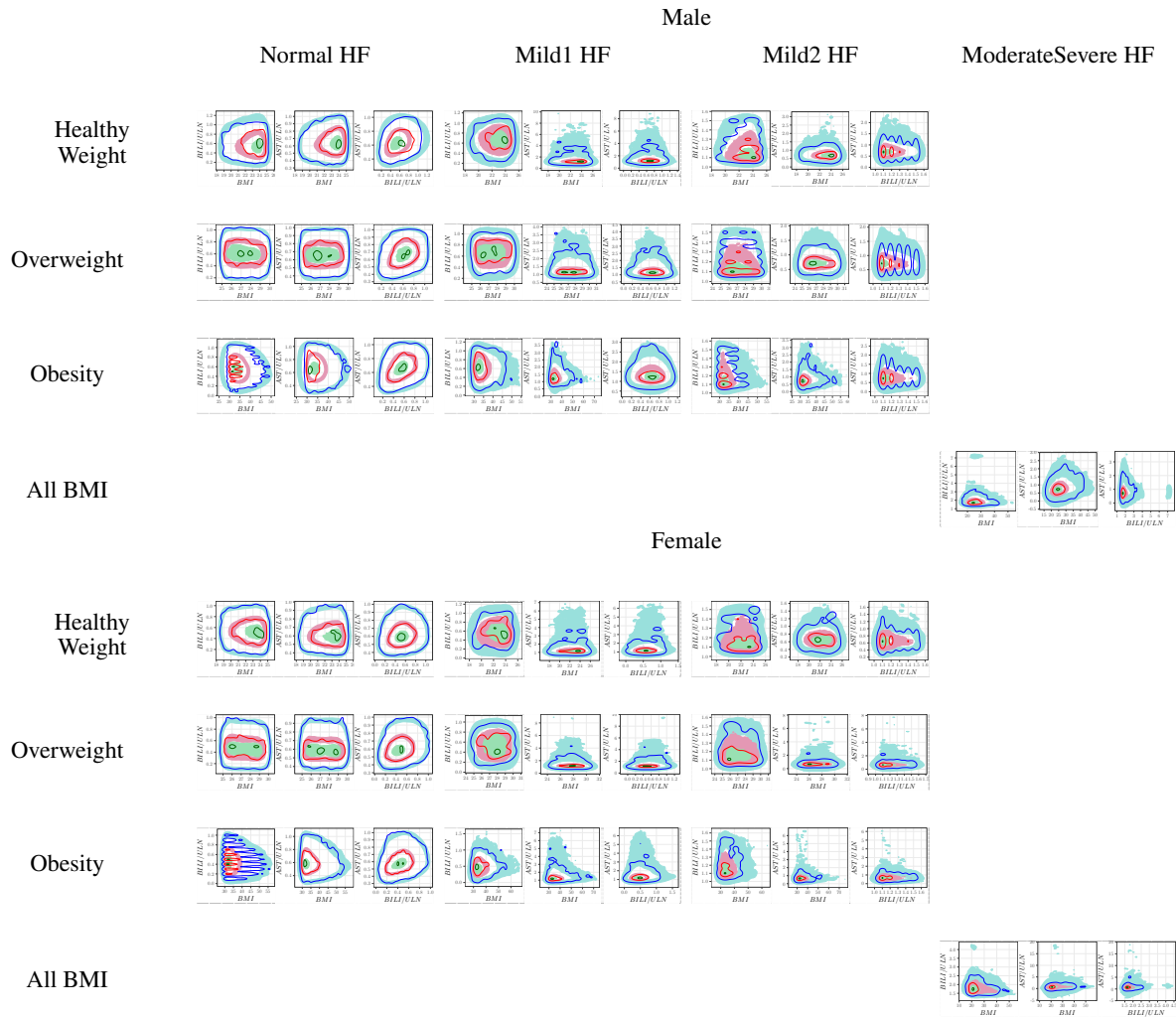


Figure 3: Hepatic case study - Visual predictive checks based on the contours of the bivariate density between *BMI*, *BILI/ULN* and *AST/ULN*. For each of the 20 covariate combination, the virtual population was simulated 100 times from the copula and the 99% prediction intervals of percentile contours of the joint distribution was derived and compared to the contours observed in the NHANES database.

The ribbon areas correspond to the 99% prediction interval of 5th (●), 50th (●) and 95th (●) percentile contours of the joint distributions. The lines corresponds to the observed 5th (—), 50th (—) and 95th (—) percentile contours.

4.2. FIM evaluation using quadrature

4.2.1. Renal case study

Figure 4 presents the convergence in relative difference of the D-criterion. The relative difference in ϕ_D between two successive number of nodes dropped below 0.5% at $n^{(Q)} = 5$, yielding a D-criterion value of 258.60. Using Monte Carlo integration (Figure 5) with $n^{(MC)} = 500$ repeated 10 times, the target D-criterion was 260.1 (sd = 0.8), corresponding to a difference of 0.6% with the GLQ of 5 nodes. However, GLQ required $n^{(Q)^2} = 25$ FIM evaluations for each covariate distributions, while for Monte Carlo, the relative error compared to the target was not controlled $\pm 0.5\%$ for $n^{(MC)} = 25$ (Figure 5).

Moreover, as shown in Appendix B Figure B.19, the convergence behaviour for GLQ was consistent across the two IIV scenarios, regardless of the ratio value. Therefore, the value $n^{(Q)} = 5$ was selected for the optimisation step.

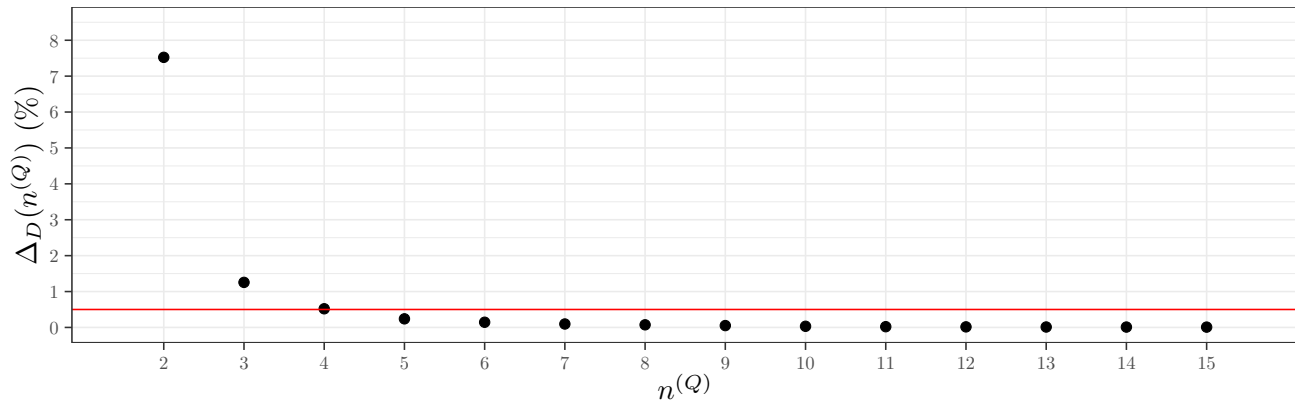


Figure 4: Renal case study - Relative difference in D-criterion as a function of the number of nodes, $n^{(Q)}$, in the Gauss-Legendre Quadrature used for Fisher Information Matrix integration

The red line corresponds to 0.5%.

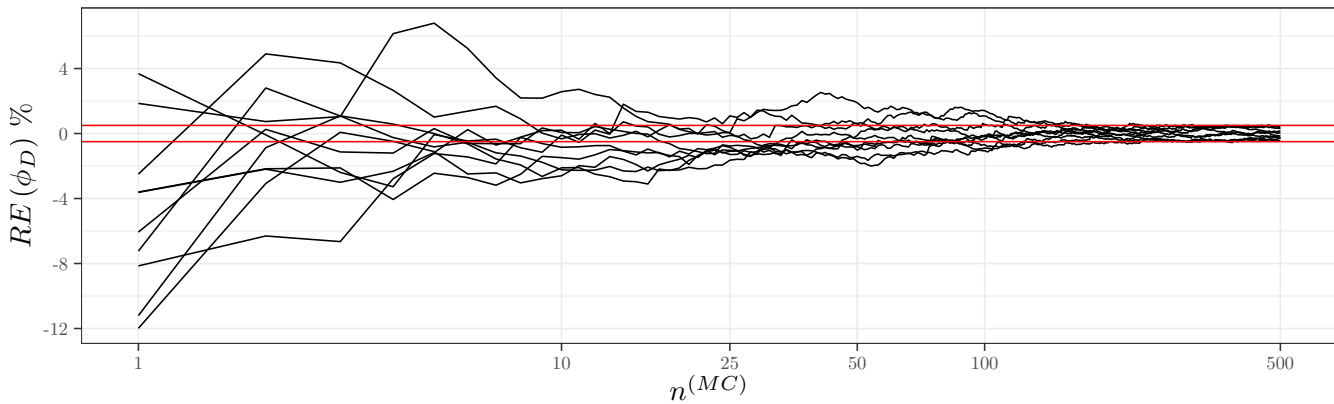


Figure 5: Renal case study - Relative error compared to the target D-criterion computed as the average across 10 repetitions of FIM integration using $n^{(MC)} = 500$ Monte Carlo samples as a function of the number of Monte Carlo samples used in Fisher Information Matrix integration

The red line corresponds to the target $\pm 5\%$.

4.2.2. Hepatic case study

In the HF case study, the relative difference in the overall D-criterion also dropped below 0.5% at $n^{(Q)} = 6$, yielding a D-criterion value of 240.1 (Figure 6). Using Monte Carlo integration (Figure 7) with $n^{(MC)} = 500$ repeated 10 times, the target D-criterion was 243.8 (sd = 0.9), corresponding to a difference of 1.5%, deemed acceptable and $n^{(Q)} = 6$ was kept for optimisation. Of, note, the convergence was not monotonic in this case study, but remained controlled under a difference of 0.5% from $n^{(Q)} = 6$.

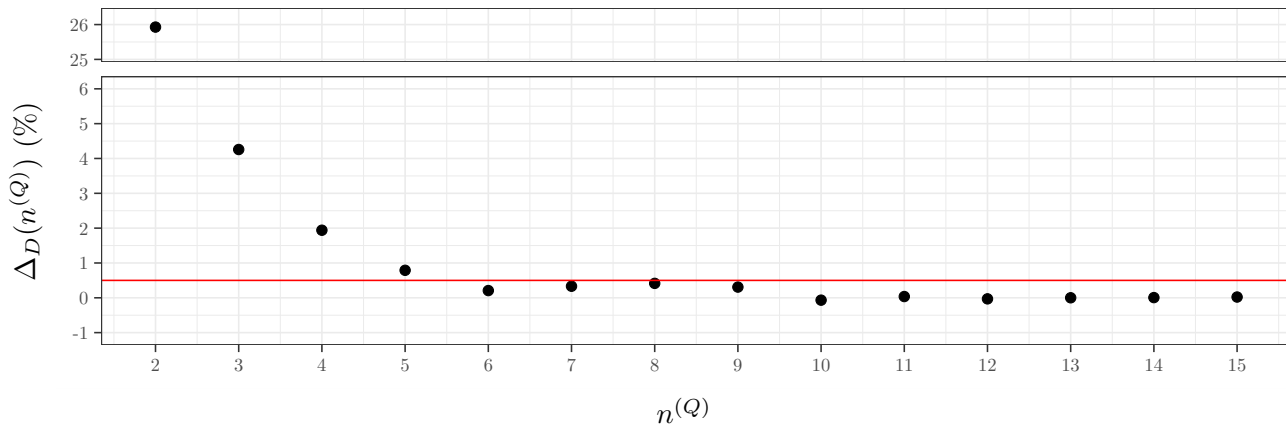


Figure 6: Hepatic case study - Relative difference in D-criterion as a function of the number of nodes, $n^{(Q)}$, in the Gauss-Legendre Quadrature used for Fisher Information Matrix integration

The red line corresponds to the 0.5%.

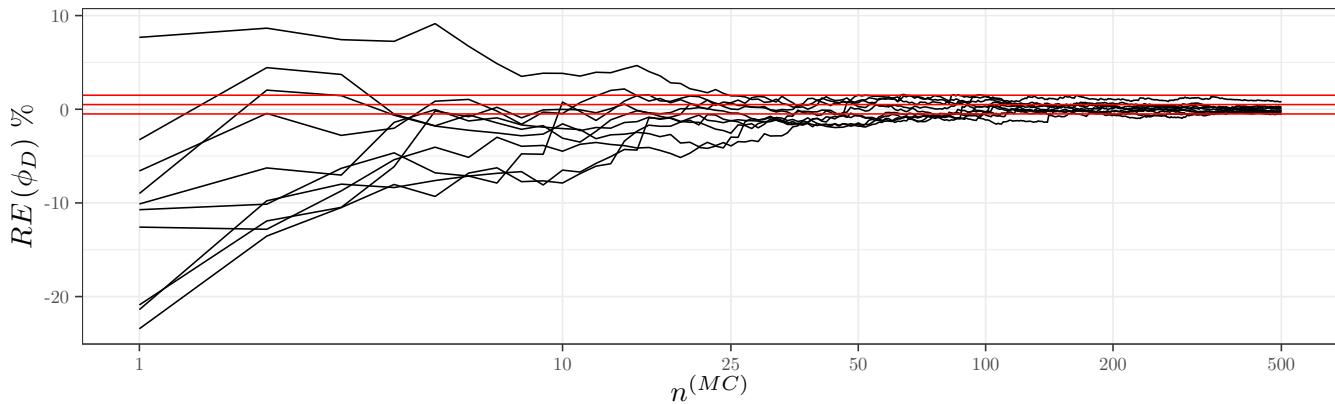


Figure 7: Hepatic case study - Relative error compared to the target D-criterion computed as the average across 10 repetitions of FIM integration using $n^{(MC)} = 500$ Monte Carlo samples as a function of the number of Monte Carlo samples used in Fisher Information Matrix integration

The red line corresponds to the target $\pm 5\%$.

4.3. Covariate distribution optimisation

In both case studies, the PGD converged without trouble and convergence plots are given in Appendix C, Figures C.20 and C.22.

4.3.1. Renal case study

The distribution between the 20 covariate combinations in the RF case study are given in Figure 8 for the initial distribution and for the two optimisation settings. The first observation is that the optimal distributions were very sparse in the sense that many categories had a weight of 0.

Because being a Female reduces V while V increases with the BMI , in optimal distributions, Female were more associated with Healthy Weight while Male were associated with Obesity. Similarly, because having a Severe RF decreases Cl and therefore slows the elimination and leads to higher concentrations, in optimal distributions, only Female Severe RF were present.

Overall, the general shape of the optimal distributions remained relatively stable across different levels of IIV and $CLCR$ effect (Appendix C Figure C.21). However, it can be observed that with increasing IIV or decreasing covariate effect magnitude, greater weight was assigned to more renal impaired.

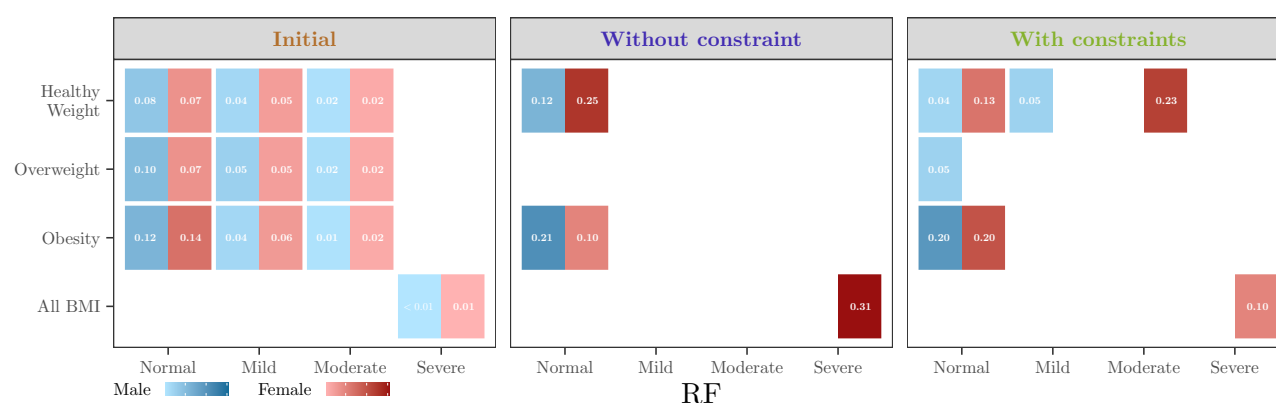


Figure 8: Renal case study - Optimal proportions for the covariate distributions for the D-criterion for the different sets of constraints

Marginal distributions are provided in Appendix C, Table C.5. For SEX , the optimal allocation consistently favoured females, with 66% representation, regardless of the constraint settings. For BMI intervals, Overweight individuals appeared in the optimal distribution, even without explicit constraints, due to the pooling of BMI categories within the Severe RF combinations. The proportions of Healthy Weight and Obesity were nearly unchanged between the unconstrained and constrained scenarios, 47% and 43% without constraints, versus 49% and 43% with constraints, respectively. Regarding RF, without constraint, Mild and Moderate RF were absent from the optimal distribution, resulting in a Normal–Severe RF split of 69% and 31%. With constraints, Mild RF was included at its minimum allowable level, and due to the additional upper constraint, Severe RF were at 10% and 23% were reallocated to Moderate RF.

Regarding performance, both optimisation settings improved D-efficiency (Table 3), by 23% without constraints and by 19% with constraints. Even under this more realistic constrained setting, the NSN was substantially reduced. Specifically, the highest NSN required to conclude on the relevance of being Female on V , of having Severe RF on Cl , and the non-relevance of Obesity on V was 602 with the initial design. This number dropped to respectively 197 and 230 without and with constraints. Thus, even the more constrained optimisation reduced the NSN by over 60%. The corresponding forest plot is shown in Figure 9 where it is evident that the initial distribution would not have allowed for a conclusion on relevance.

Regarding sensitivity analysis, as expected the greater the effect of RF (i.e. the smaller the

Table 3: Renal case study - D-criterion, D-Efficiency and Number of subjects needed (NSN) to reach 80% power in relevance or non-relevance tests, with the initial and optimal covariate distributions for the D-criterion for the different sets of constraints

Name	Initial Design	Optimal designs	
		Without constraint	With constraints
D-criterion	258.6	319.0	308.8
D-Efficiency	1.00	1.23	1.19
$NSN_R(V, \text{Female})$	167	197	196
$NSN_{NR}(V, \text{Obesity})$	143	118	107
$NSN_R(Cl, \text{Severe RF})$	602	143	230

in bold: the higher NSN across the three covariate relationships, for respectively the significance tests and the (non-)relevance tests, for each design.

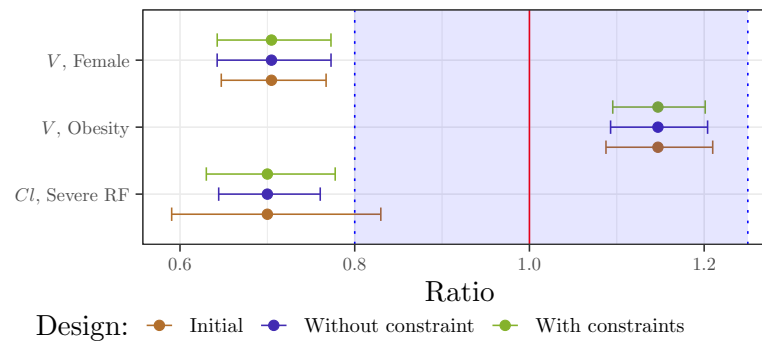


Figure 9: Renal case study - Ratios of covariate effects and their 90% confidence interval predicted with the Fisher Information Matrix for $N = 230$ subjects, using the initial covariate distribution and the optimal ones, without and with constraints.

The red line at 1 corresponds to the reference line i.e., no change from the typical individual; the shaded area in blue represents the reference area of $[0.80, 1.25]$.

ratio), the smaller the NSN to conclude on relevance of Severe RF effect (Figure 10). For a same effect magnitude, the NSN increased with IIV.

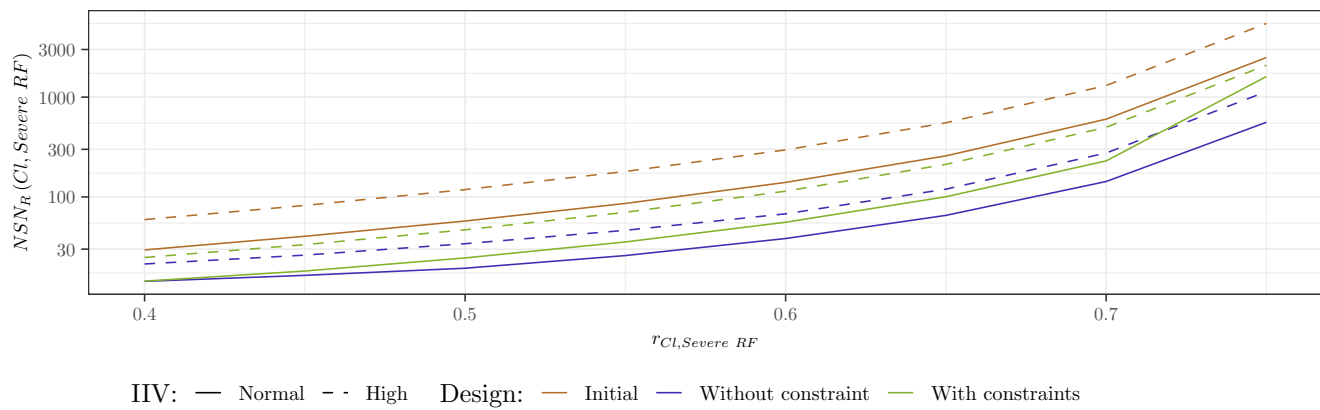


Figure 10: Renal case study - $NSNR(Cl, CLCR \text{ (Severe RF)})$ to reach 80% power in relevance test (right) for the two IIV scenarios and as a function of $r_{Cl, \text{Severe RF}}$ from 0.40 to 0.75

4.3.2. Hepatic case study

For the HF case study, the optimal distributions are shown in Figure 11. As in RF case study, the optimal allocation primarily weighted extreme values, with no Overweight individuals included in the unconstrained setting. As above, because ModerateSevere HF decreases Cl , Females were more associated with Healthy Weight and ModerateSevere HF, while Males were associated with Obesity and Normal HF. Notably, the optimal distributions included a higher proportion (72%) of Females than Males (Appendix C, Table C.6). Similarly to the RF case study, the unconstrained optimal marginal distribution for HF was mainly concentrated between Normal HF (44%) and ModerateSevere HF (40%). In the constrained setting, where a 10% upper bound was imposed on ModerateSevere HF, part of this group was replaced by Mild HF which became the majority.

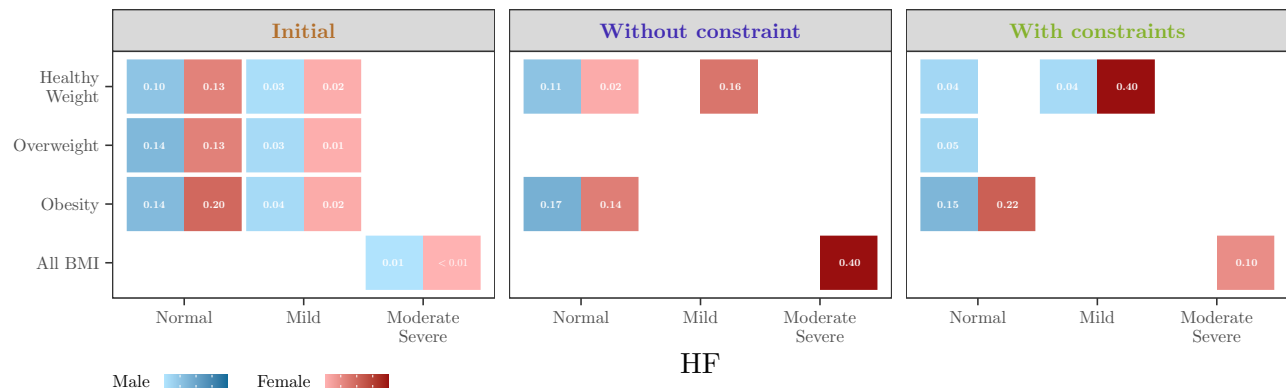


Figure 11: Hepatic case study - Optimal proportions for the covariate distributions for the D-criterion for the different sets of constraints

Regarding performances, optimisation once again increased the D-criterion and reduced the NSN (Table 4). The NSN to conclude on the relevance of the effect of being a Female on V and of the effect of having ModerateSevere HF on Cl , and on the non-relevance of having Obesity on V was 400 with the initial design. This number dropped to 216 with the constrained optimisation, representing a reduction of 46%. The corresponding forest plot is shown in Figure 12.

Table 4: Hepatic case study - D-criterion, D-Efficiency and Number of subjects needed (NSN) to reach 80% power in relevance or non-relevance tests, with the initial and optimal covariate distributions for the D-criterion for the different sets of constraints

Name	Initial Design	Optimal designs	
		Without constraint	With constraints
D-criterion	240.1	304.0	294.4
D-Efficiency	1.00	1.27	1.23
$NSN_R(V, SEX)$	167	217	216
$NSN_{NR}(V, Obesity)$	140	105	108
$NSN_{NR}(Cl, ModerateSevere HF)$	400	148	141

in bold: the higher NSN across the three covariate relationships, for respectively the significance tests and the (non-)relevance tests, for each design.

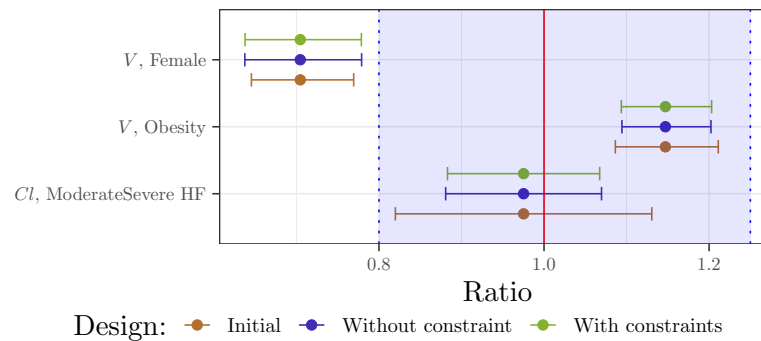


Figure 12: Hepatic case study - Ratios of covariate effects and their 90% confidence interval predicted with the Fisher Information Matrix for $N = 216$ subjects, using the initial covariate distribution and the optimal ones, without and with constraints.

The red line at 1 corresponds to the reference line i.e., no change from the typical individual; the shaded area in blue represents the reference area of $[0.80, 1.25]$.

5. Discussion

In this work we proposed to optimise the allocation of the covariates among the patients to be including in a future population PK study to assess the relevance or non-relevance of covariate relationships using NLMEM. We applied the proposed workflow to a PK model through two case studies, assessing the effects of either renal or hepatic impairment. We have first introduced a new way of integrating the FIM for the covariate distribution by leveraging copula modelling and GLQ. The optimisation was carried out between categories for discrete covariates, while for continuous ones, we introduced a segmentation of their domains into intervals and optimised the proportion between them. We solved the optimisation problem using the PGD algorithm. All these methods have been implemented in R based on the PFIM 6.1 package, and the source code is available online at the following link: <https://doi.org/10.5281/zenodo.14778033>.

In our example we used a covariate database on which copula were fitted as initial distributions. Goodness of fit diagnosis suggested that a large amount of data is required for estimating copula but the latter could have been used directly if available, which is one of their great interests in pharmaceutical applications as they can be made public without threatening anonymity. In addition, the optimisation approach can be coupled with another method for FIM computation if preferred. Of note, in the HF case study, we chose to fit separately the Mild1 and and the Mild2 data, although diagonal copula have been studied in the literature (see, for instance, (de Amo

et al., 2016; Bukovšek et al., 2024)), but this topic was beyond the scope of our work.

We have shown that using GLQ for FIM integration with respect to covariate distribution provided good accuracy compared to Monte Carlo, while reducing the number of FIM evaluations and thus, computation time. We proposed a practical approach for users to determine the required number of quadrature nodes by monitoring the relative change in the D-criterion. If greater precision is required, the number of quadrature nodes can be increased; however, doing so may undermine the computational efficiency that motivates the use of quadrature methods compared to Monte Carlo. Indeed, the GLQ approach is limited by dimensionality: as the number of continuous covariates C increases, the number of FIM evaluations grows exponentially as $(n^{(Q)})^C$. This challenge could be mitigated by refining the number of nodes per covariate, rather than using a uniform value or by employing adaptive quadrature methods that require fewer nodes (see, for example, the review in (Malcolm and Simpson, 1975)). That said, in practice it is unlikely that a large number of covariates would be optimized simultaneously as the objective is typically to evaluate only a few key covariates, as illustrated in our example. Moreover, GLQ remains computationally more efficient than CTS, which requires simulating and estimating a large number of datasets. If quadrature becomes computationally prohibitive, quasi-Monte Carlo methods (Pan and Thompson, 2007) may offer a practical compromise between standard Monte Carlo integration and deterministic quadrature, balancing accuracy and efficiency.

The introduced covariate segmentation enables a fast optimisation without the need to recompute FIM. This allows for the exploration of various constraints and optimisation settings without concern for computational cost. This optimisation was performed using the PGD with a absolute tolerance fixed at $\epsilon = 10^{-5}$ and a fixed step size. We did not explore these PGD hyper-parameters nor some of performance-improving variants because the need was not felt in our situation, but it can be an option in more complex cases. Especially, adaptive step size (Duchi et al., 2011) may have improved the solution or the computing time, which could be useful with higher dimensionality. In the current version of this algorithm, only linear constraints can be used for an easy projection on the constraint space, while in practice some non-linear ones may be useful (e.g., to impose the power of a test or a SE level on a parameter). To address this, a penalty term could be added to the objective function or another class of algorithms could be explored such as interior-point method (Lu and Pong, 2013). Regarding the optimisation criteria, we used the D-criterion and others exist such as the DS-or DDS-criterion which allow focus on a specific subset of the parameters (see for instance (Seurat et al., 2020) for an application). In our case, the covariates effects could have been the only parameters of interest. This change of criterion is not straightforward in the sense that its derivative has to be calculated, but apart from this calculation, it presents no particular difficulty with the PGD algorithm. The resulting optimal distributions may not correspond to integer subject counts within subgroups; however, efficient rounding methods (Pukelsheim and Rieder, 1992) can be applied to convert them into realistic sample sizes.

Regarding FIM computation, we used a linearisation approach but methods avoiding this approximation have already been explored using Monte Carlo and adaptive GQ (Riviere et al., 2016) (Ueckert and Mentré, 2017). A future work could explore a full evaluation of the FIM with no linearisation and covariate integration using these approaches. It would also allow extension to non-continuous response, for which the FIM cannot be computed with linearisation.

Last but not least, this approach assumes that the model and its parameters are known, including the effects of the covariates, whereas the trial to be designed aims to better estimate them, and is not robust to misspecifications. In the RF case study, we computed optimal allocation and NSN for a range of β and this could be extended into a robust approach as already explored

in the NLMEM context (Dodds et al., 2005; Foo et al., 2012; Loingeville et al., 2020; Seurat et al., 2020). Moreover, this methodology could be used to design a Phase 3 study based on Phase 2 results; therefore, the SE from the covariate effects estimates based on the Phase 2 data may be used to specify a prior distribution on the β . This absence of robustness also holds for the covariate distribution assumed for FIM computation and covariate optimisation. In this case, Adaptive Designs and especially Two-stage approaches (Dodds et al., 2005; Lestini et al., 2015; Fayette et al., 2023) could be a good avenue to explore for refining the distribution of covariates halfway through inclusion.

From an application point of view, we illustrated our method through a PK example with two case studies in which we optimised the allocation of the covariates among the patients to be including in clinical trials to assess the relevance and non-relevance of renal and hepatic impairment. We especially focused on *NSN* to conclude on the relevance or non-relevance of the associated covariates with a sufficient degree of confidence. For this assessment, we used the equivalence region $[0.80; 1.25]$ which is the one used in bioequivalence studies to define whether a dosing adjustment is required but alternative methods have been proposed to choose equivalence regions appropriate to a specific context (Sanghavi et al., 2024). In this work, we have assumed the same elementary design for all the subjects; the computations can easily be extended to the case where there are several. It would therefore certainly be more beneficial to adapt the elementary design to the covariate combination. Then, an interesting path may be to jointly optimize the elementary designs and the covariates. If discrete optimisation is performed for elementary designs, it can be combined with the covariates and the global process remains the same, only the number of FIM evaluations increases. We can imagine that this joint optimisation would enable us to improve the designs, for example in our case, for subjects with poor elimination, having a greater final observation time would guarantee a better estimation. On the other hand, for continuous optimisation, the extension is harder. In both cases, it will first be necessary to consider the practical and ethical constraints and discuss the actual implementation with experts. The resulting optimal distributions were consistent, typically favouring allocation to the more extreme classes, with intermediate values disappearing in scenarios without constraints. Constraints were introduced to better reflect real-world conditions for patient inclusion in clinical studies. Even with these constraints, our two case studies demonstrated that the optimal distribution greatly reduced the required *NSN* to conclude on the clinical relevance (or non-relevance) of the covariate relationships. The main challenge in implementing this optimal distribution lies in the need to recruit a substantial number of Female subjects with either Severe RF or ModerateSevere HF. Nevertheless, given the significant reduction in *NSN*, focusing efforts on these difficult-to-recruit subgroups may be justified. Moreover, constraints can be added to correspond practical settings. For instance we could have specified that we want as many Males with Severe RF (or ModerateSevere HF) as Females, or constraint each covariate combination to represent at least 5%. In addition, optimal proportions can be adapted to match actual conditions, but in this case, the *NSN* must be reevaluated using the new proportions.

Finally we introduced a flexible optimisation workflow to help the subject inclusion in clinical trials which aims to assess the relevance of covariates.

Code availability

All the R codes needed to reproduce the example in full are available online at: <https://doi.org/10.5281/zenodo.14778033>.

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830 Appendix A. Complementary results on copula diagnostics

831 Appendix A.1. Renal case study

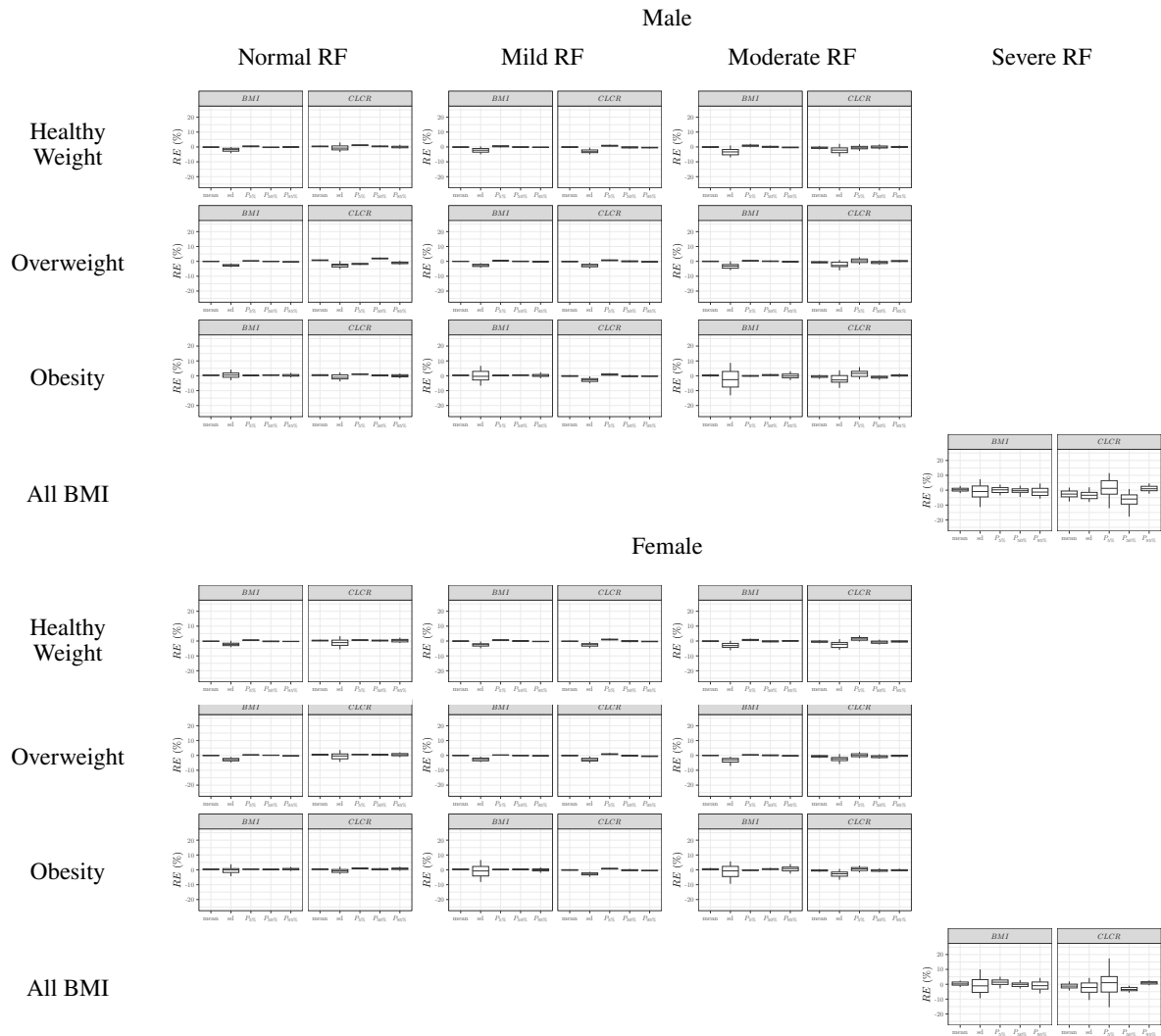


Figure A.13: Renal case study - Boxplot of the relative error (RE) of marginal distributions (mean, standard deviation and percentiles) of continuous covariates for the 20 covariate combinations, as compared to the statistics of the NHANES database. For each combination, the virtual population was simulated 100 times from the copula. The boxplot displays the median, the 25th and 75th percentiles, while the whiskers are 5th and 95th percentiles. sd refers to the standard deviation, and P5%, P50% and P95% to the 5th, 50th and 95th percentiles respectively.

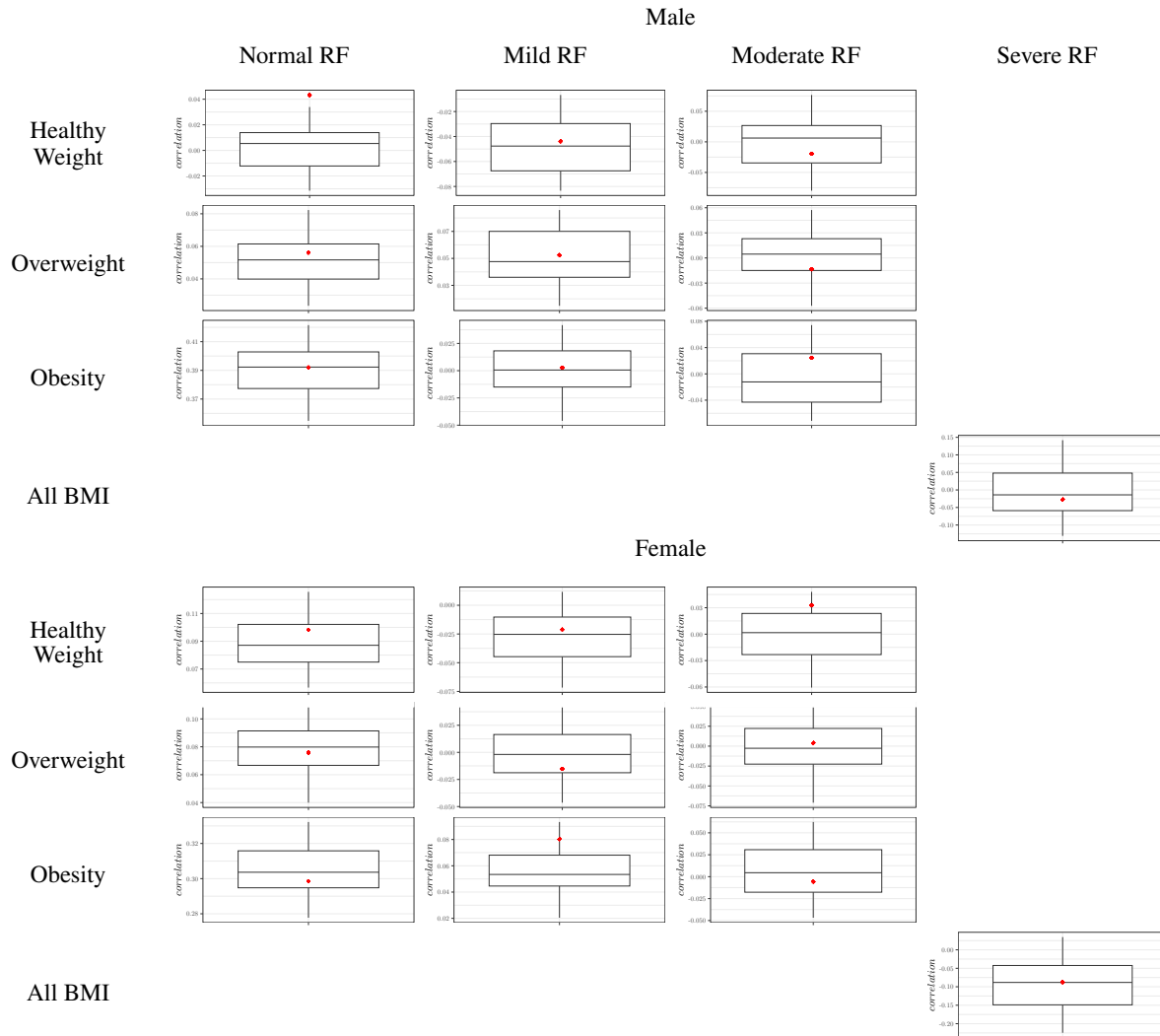


Figure A.14: Renal case study - Correlation between *BMI* and *CLCR* for each of the 20 covariate combinations, in virtual population simulated 100 times from the copula (boxplot) and in the NHANES database (red diamond). The boxplot displays the median, the 25th and 75th percentiles, while the whiskers are 5th and 95th percentiles. sd refers to the standard deviation, and P5% , P50% and P95% to the 5th, 50th and 95th percentiles respectively.

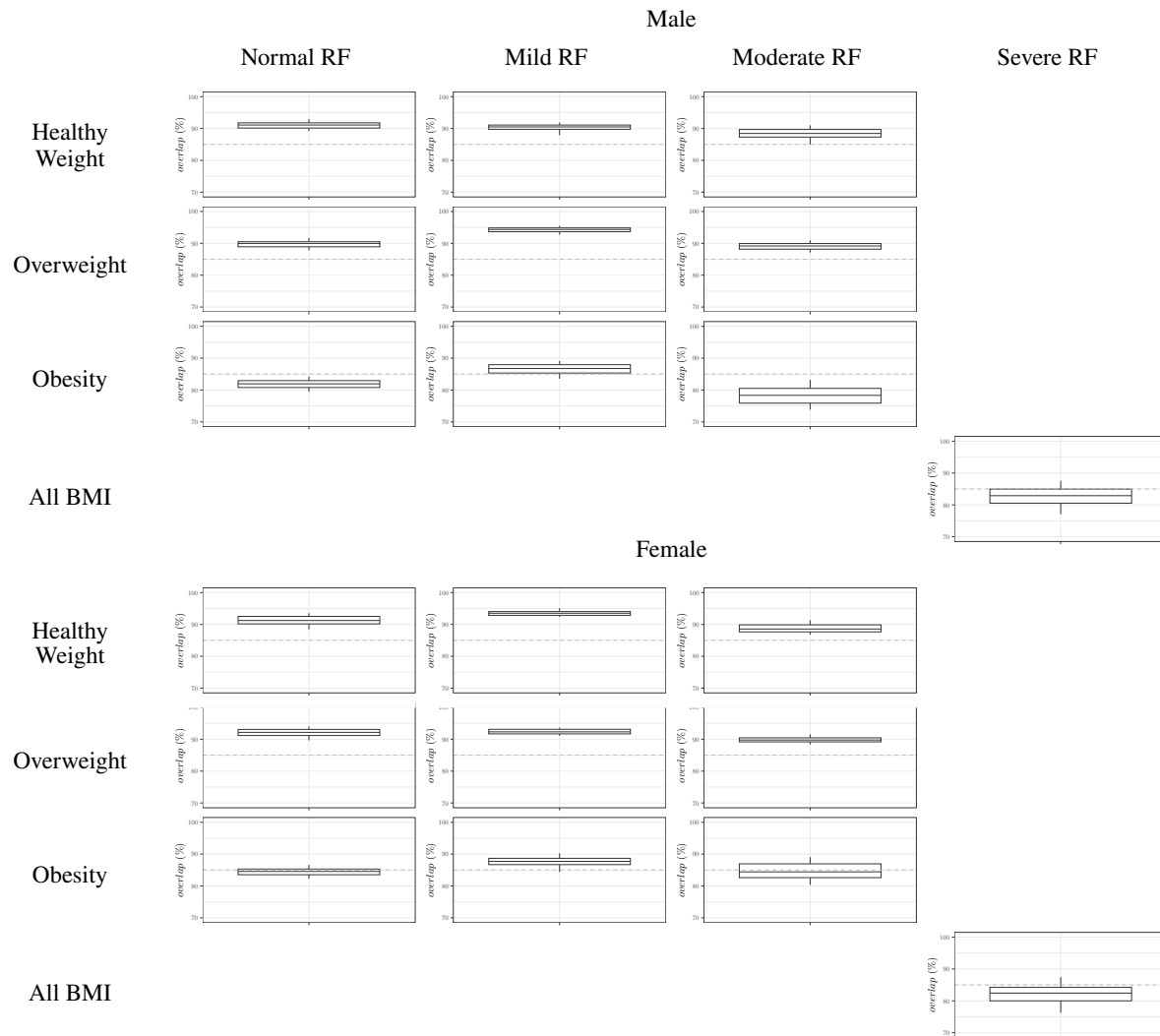


Figure A.15: Renal case study - Overlap metric between the 95th density contours of the joint distribution between *BMI* and *CLCR* in the virtual population simulated 100 times from the copula and in the NHANES database. Overlap metric of 95th density contours of virtual population relative to observed population. The gray dashed line indicates 85% overlap percentage.

832 *Appendix A.2. Hepatic case study*

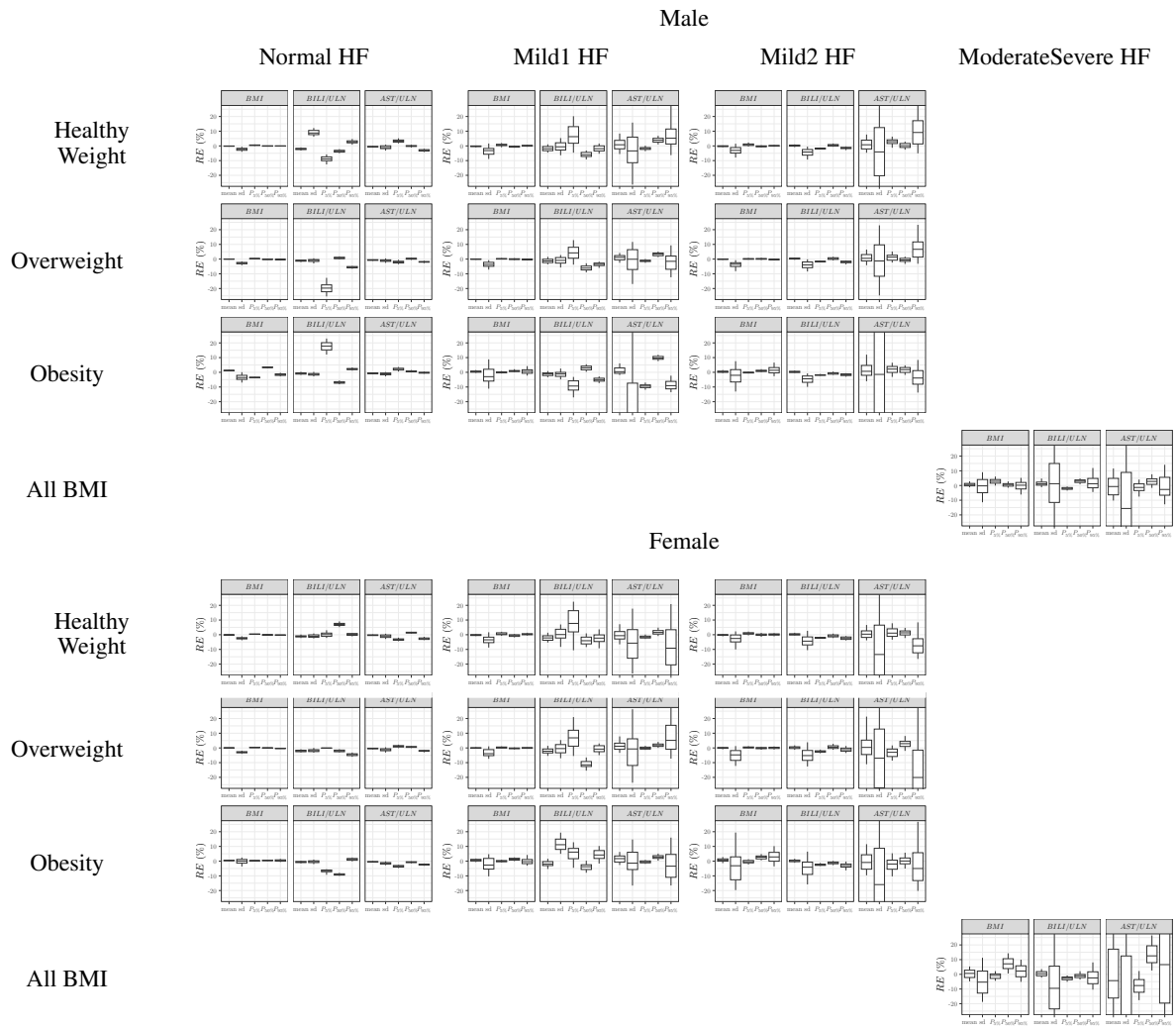


Figure A.16: Hepatic case study - Boxplot of the relative error (RE) of marginal distributions (mean, standard deviation and percentiles) of continuous covariates for the 18 covariate combinations, as compared to the statistics of the NHANES database. For each combination, the virtual population was simulated 100 times from the copula. The boxplot displays the median, the 25th and 75th percentiles, while the whiskers are 5th and 95th percentiles. sd refers to the standard deviation, and $P5\%$, $P50\%$ and $P95\%$ to the 5th, 50th and 95th percentiles respectively.

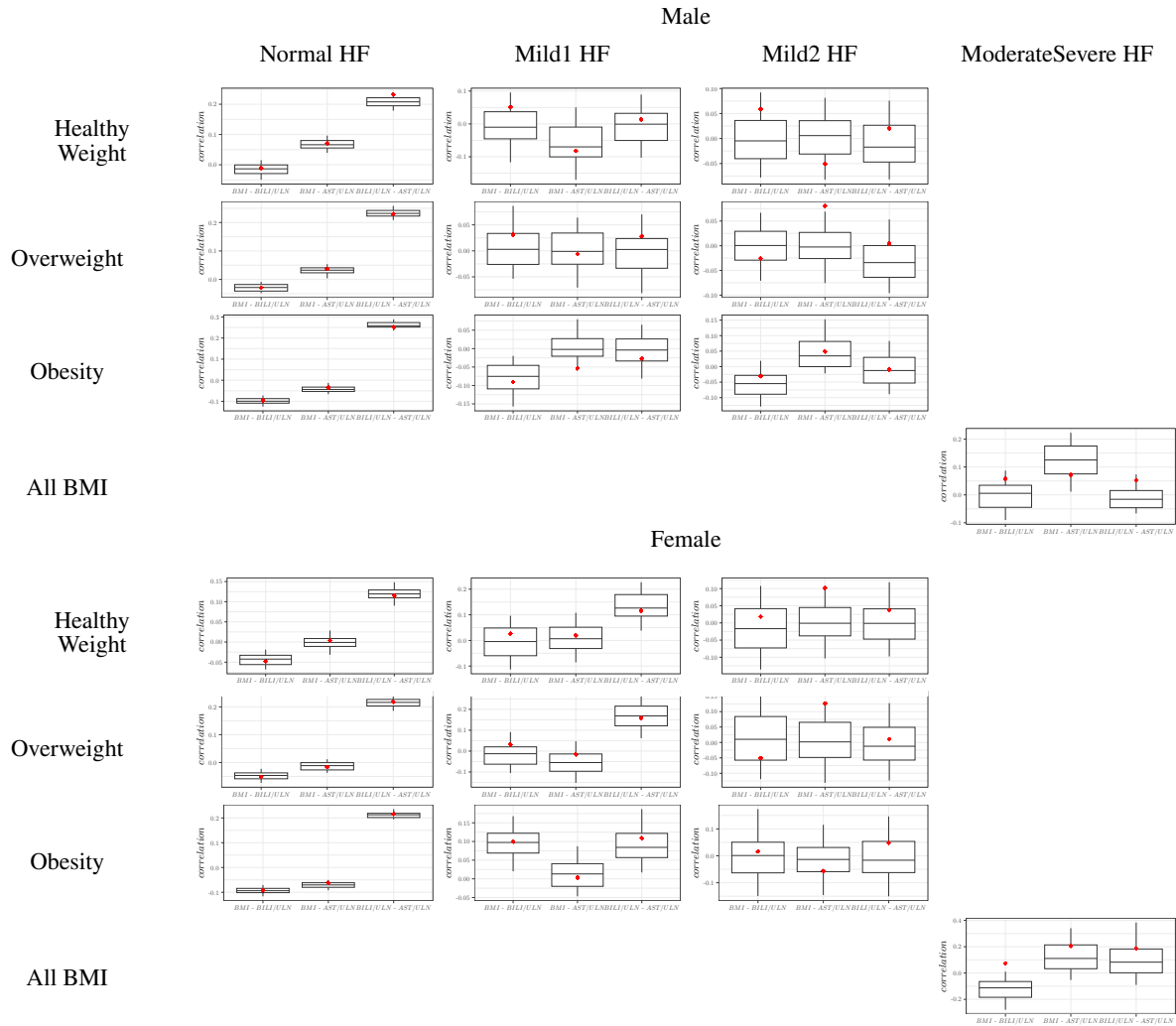


Figure A.17: Hepatic case study - Correlations between BMI , $BILI/ULN$ and AST/ULN for each of the 18 covariate combinations, in virtual population simulated 100 times from the copula (boxplot) and in the NHANES database (red diamond).

The boxplot displays the median, the 25th and 75th percentiles, while the whiskers are 5th and 95th percentiles. sd refers to the standard deviation, and P5%, P50% and P95% to the 5th, 50th and 95th percentiles respectively.

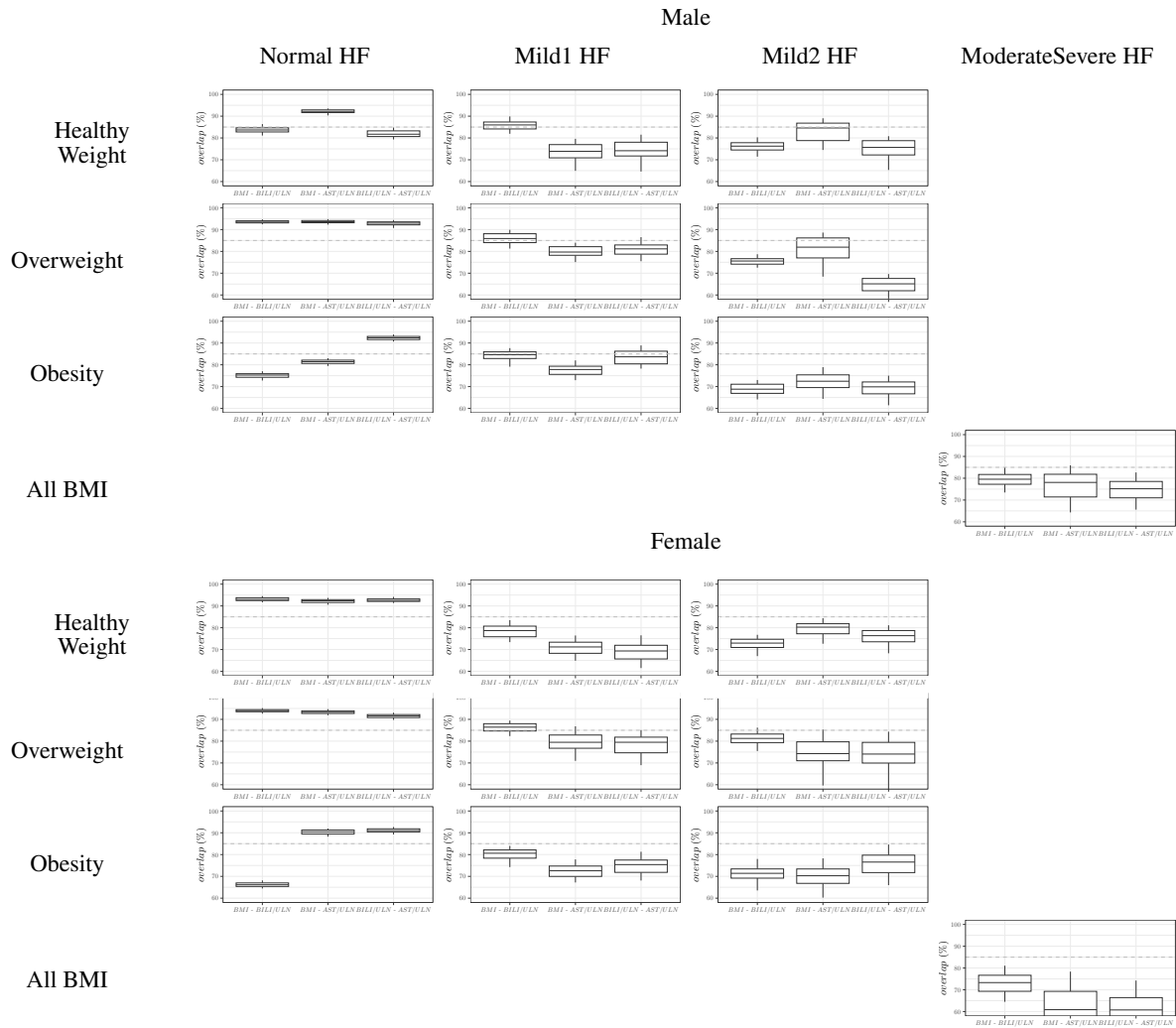


Figure A.18: Hepatic case study - Overlap metric between the 95th density contours of the joint distributions between *BMI*, *BILI/ULN* and *AST/ULN* in the virtual population simulated 100 times from the copula and in the NHANES database.

Overlap metric of 95th density contours of virtual population relative to observed population. The gray dashed line indicates 85% overlap percentage.

833 Appendix B. Complementary results on FIM evaluation using quadrature

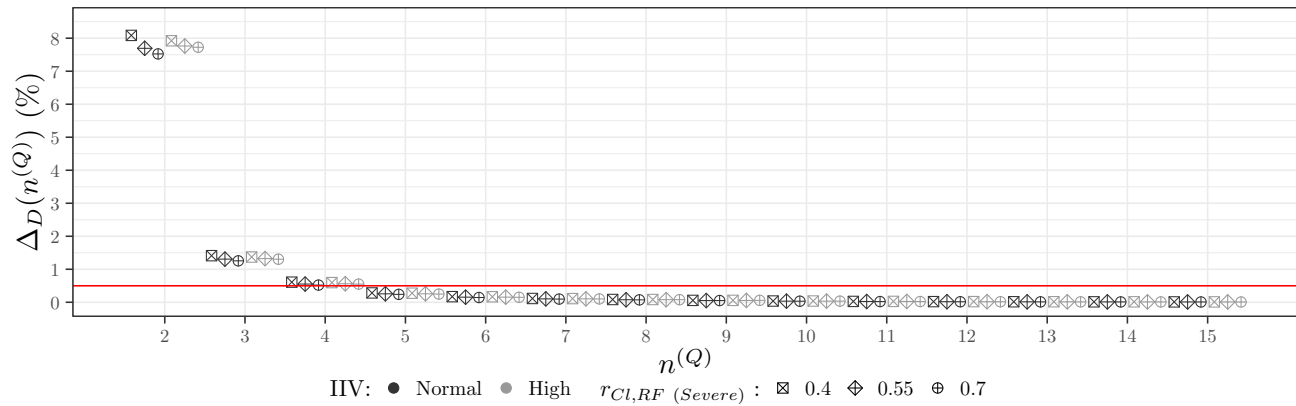


Figure B.19: Renal case study - Sensitivity analysis for the relative difference in D-criterion as a function of the number of nodes, $n^{(Q)}$, in the Gauss-Legendre Quadrature used for Fisher Information Matrix integration, for the two IIV scenarios and three values of $r_{CI, RF} (Severe)$. The red lines corresponds to the 0.5% target.

834 Appendix C. Complementary results on covariate distribution optimisation

835 Appendix C.1. Renal case study

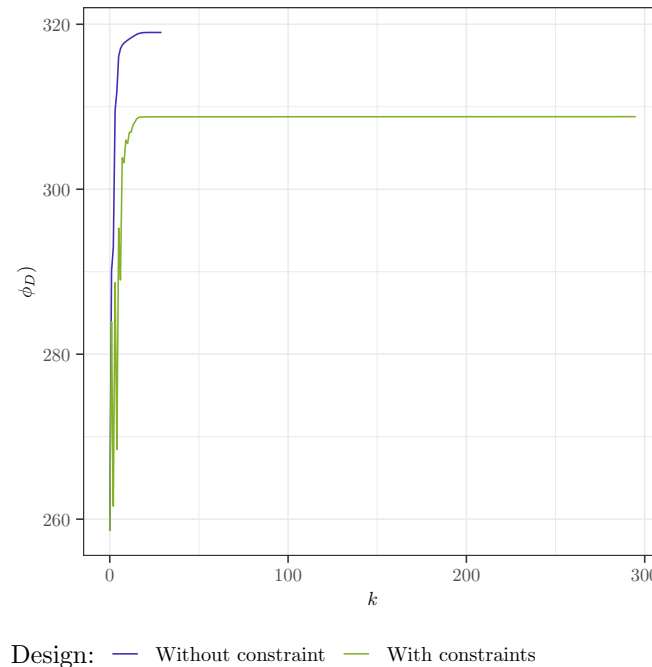


Figure C.20: Renal case study - D-criterion as a function of the iterations in the Projected Gradient Descent optimisation for the different sets of constraints

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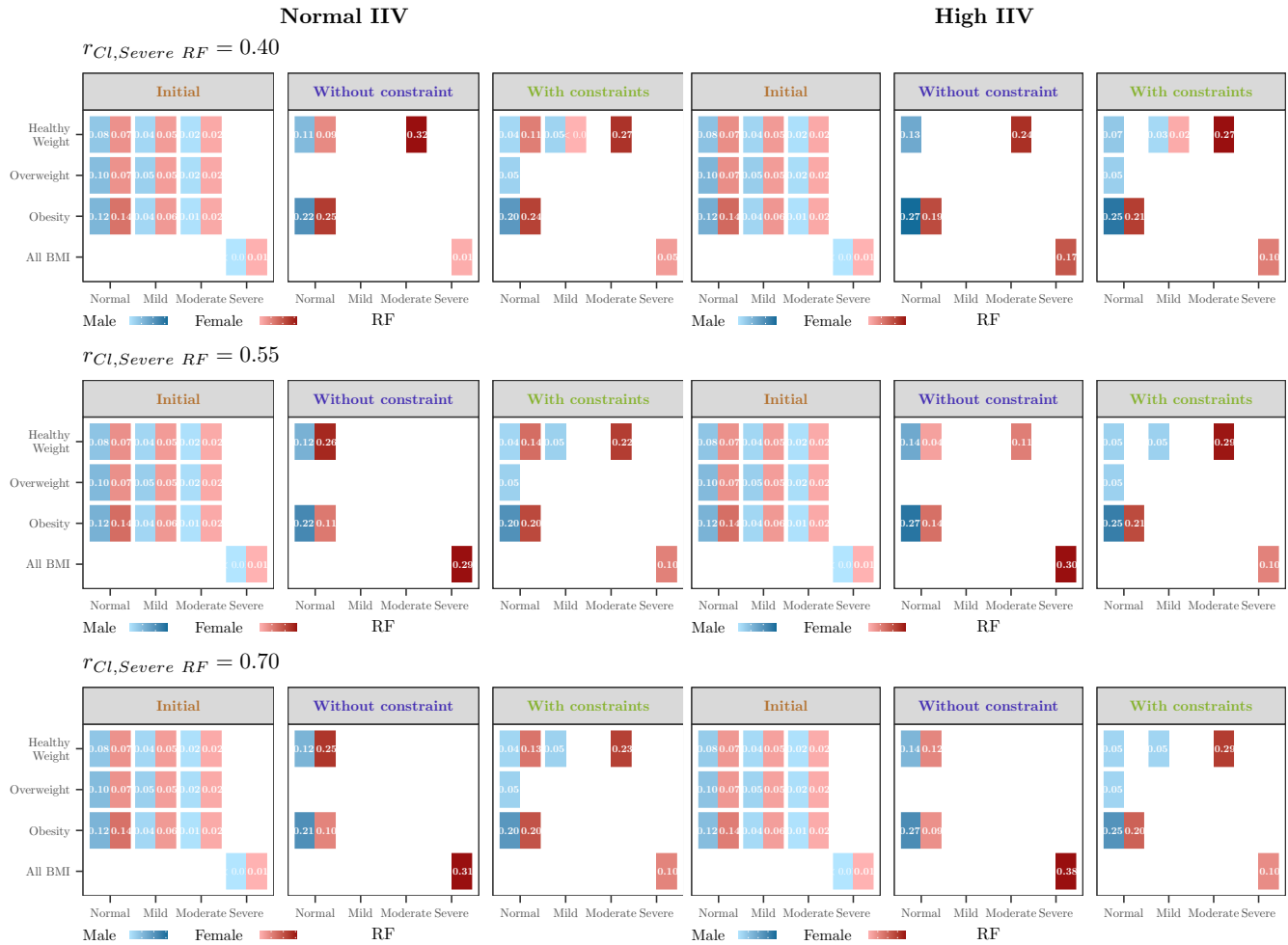


Figure C.21: Renal case study - Optimal proportions for the covariate distributions for the D-criterion for the different sets of constraints, for the scenario with Normal IIV (left) and High IIV (right) and for three value of $r_{CI, Severe RF}$.

Table C.5: Renal case study - Marginal proportions for the covariate distributions for the D-criterion for the different sets of constraints

Covariate	Interval	Initial	Without constraint	With constraints
<i>SEX</i>	Male	0.49	0.34	0.34
	Female	0.51	0.66	0.66
<i>BMI</i>	Healthy Weight	0.28	0.47	0.49
	Overweight	0.33	0.10	0.08
	Obesity	0.39	0.43	0.43
RF	Normal	0.58	0.69	0.62
	Mild	0.28	0.00	0.05
	Moderate	0.12	0.00	0.23
	Severe	0.01	0.31	0.10

836 *Appendix C.2. Hepatic case study*

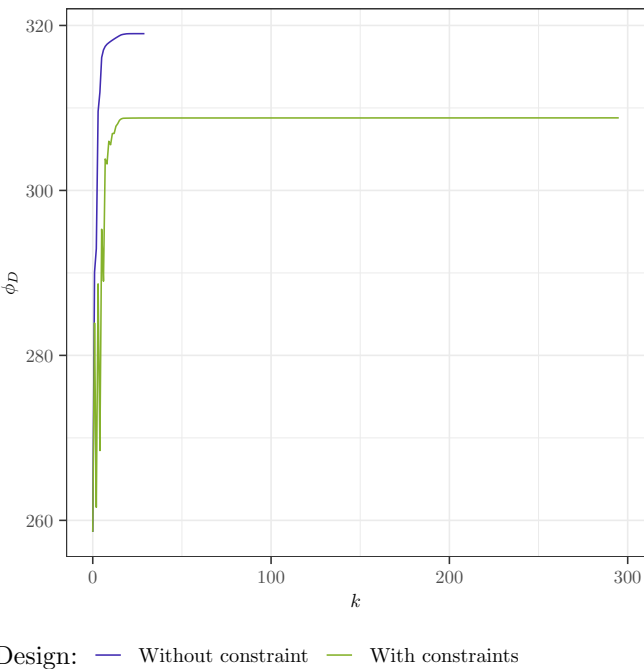


Figure C.22: Hepatic case study - D-criterion as a function of the iterations in the Projected Gradient Descent optimisation for the different sets of constraints

Table C.6: Hepatic case study - Marginal proportions for the covariate distributions for the D-criterion for the different sets of constraints

Covariate	Interval	Initial	Without constraint	With constraints
<i>SEX</i>	Male	0.49	0.28	0.28
	Female	0.51	0.72	0.72
<i>BMI</i>	Healthy Weight	0.28	0.52	0.54
	Overweight	0.33	0.06	0.06
	Obesity	0.39	0.42	0.40
HF	Normal	0.84	0.44	0.46
	Mild	0.15	0.16	0.44
	ModerateSevere	0.01	0.40	0.10